Stock Analysis

Fundamental Analysis and Financial Modelling

1 How Share Prices Move/Fluctuate

Share prices are determined by the basic economic principle of supply and demand:

$$Demand \uparrow Supply \downarrow \implies Price \uparrow$$
$$Demand \downarrow Supply \uparrow \implies Price \downarrow$$

Key Driver: Information

1.1 Information Sources

Market information comes from various channels that influence investor decisions:

- Annual Reports
- Quarterly Earnings
- Court Cases
- Press Releases
- News Stories
- Social Media
- Market Buzz

1.2 Primary Influencing Factors

Share price movements are influenced by four major categories of factors:

1. Company Fundamentals

The fundamental strength of a company forms the foundation of its valuation:

- Earnings
- Growth Prospects
- Management & Governance
- Competitive Advantage
- Expansion Plans
- Debt Position
- Promoter Shareholding
- Pledging of Shares

2. Technical Indicators

Technical factors reflect market sentiment and trading dynamics:

- Liquidity
- Positive/Negative News Flow
- Chart Patterns
- Market Valuation (Overpriced/Underpriced)

3. Micro & Macro Economic Factors

Broader economic conditions significantly impact stock performance:

- Socio-Political Stability/Instability
- Government Policies
- National & Global Economy/GDP Trends
- Inflation Rates
- Competitive Investment Landscape

4. Human Sentiment

Market psychology plays a crucial role in price movements:

- Information Asymmetry
- Divergent Reactions to a Singular Piece of Information

(Trend is your friend. Follow the trend, not news.)

1.3 Investing vs Trading

The two primary approaches to market participation differ significantly:

Investing	**Trading**
Generally for the long term	Typically for the short or very short term
Utilizes Fundamental Analysis	Relies on Technical Analysis
Associated with lower risk	Involves higher risk
Aims for wealth creation	Aims to generate profits (make money)
Requires less active time participation	Demands more active time commitment

2 Investment and Trading Strategies

2.1 Types of Investing

Growth Investing

Growth investors focus on companies with exceptional growth potential:

- Growth investors are attracted to companies that are expected to grow faster (by revenues/cash flow and profits) than the rest.
- *Father: Philip Fisher*

Value Investing

Value investors seek undervalued opportunities in the market:

- It is about finding companies whose stock price doesn't necessarily reflect their fundamental worth.
- *Father: Benjamin Graham*

2.2 Types of Trading

Different trading strategies are categorized by their time horizons:

1. **Scalp Trading**
   - Duration: Seconds, minutes; Buy-Sell
2. **Intraday Trading**
   - Duration: 1 day; Buy-Sell
3. **Swing Trading**
   - Duration: 1-7 days; based on short-term trends
4. **Momentum Trading**
   - Strategy: Buy securities that are rising and sell when they appear to have peaked
5. **Positional Trading**
   - Duration: Months; based on long-term potential of stock
6. **BTST (Buy Today Sell Tomorrow) & STBT (Sell Today Buy Tomorrow)**

2.3 Analysis Methods

Fundamental Analysis

Fundamental analysis provides the foundation for investment decisions:

- Attempts to evaluate stocks by measuring their intrinsic value.
- Involves studying everything from the overall economy and industry to the financial strength and management of individual companies.
- The assumption is that in the longer horizon, the share price of a company with good fundamentals will rise, thus providing profits to investors.
- Good for long-term investments.
- Generally less risky.
- Can be more time-consuming.

Recommended Books:

1. The Intelligent Investor
2. Security Analysis
3. Common Stocks and Uncommon Profits

Technical Analysis

Technical analysis serves as the primary tool for trading decisions:

- Attempts to identify trading opportunities by looking at statistical trends, such as movements in a stock's price and volume.
- The assumption is that all known fundamentals are already factored into the price and that history tends to repeat itself.
- Primarily used for trading.
- Generally involves higher risk.
- Can be less time-consuming compared to fundamental analysis.

Recommended Books:

1. Technical Analysis of the Financial Markets by John J. Murphy
2. Numerous resources available, including YouTube videos for learning Technical Analysis

3 Fundamental Analysis

Fundamental analysis involves evaluating a company's intrinsic value through three layers of examination:

3.1 Economy Analysis

When people make transcations (buy-sell product and services) it affects the economy of a country. We need to perform economy analysis to get an idea for right time and industry to invest in.

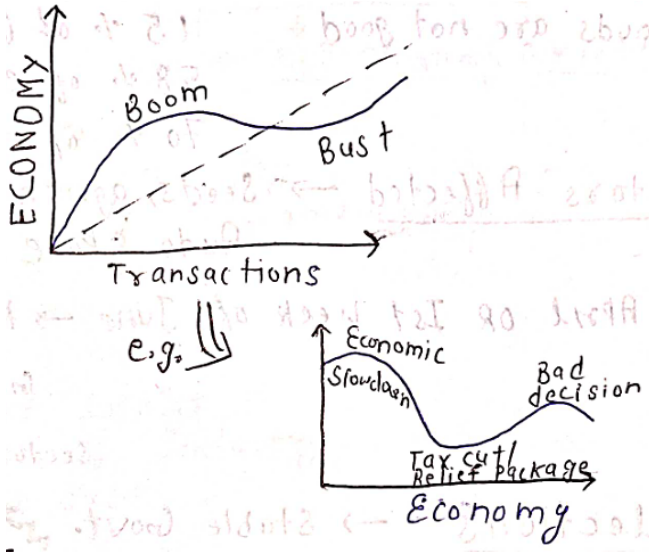

Figure 1: Economic movement

Understanding macroeconomic conditions that impact investment decisions:

- **Crude Oil Prices**
  - Increase → Economic Health ↓, Inflation ↑
  - Decrease → Good Economic Health ↔ (for India)
  - Impact: For every \$10 increase → GDP decreases by 0.4%
  - Affected sectors: Exploration & Production, Refining & Distribution, Lubricants, Chemicals, Diesel, Petrol, Natural gas, Jet fuel, LPG, Waxes, Bitumen for roads and roofing, Polishes, fuels for ships, factories like Plastic, medicine, rubber, etc.
- **GDP (Gross Domestic Product)**
  - Represents total production value of a nation
  - Higher values generally better, but lower GDP may present buying opportunities
- **Inflation**
  - Measured by Consumer Price Index (CPI) & Wholesale Price Index (WPI)
  - Lower inflation preferred for economic stability
- **Monsoons**
  - 4 months in a year accounts for 75% of annual rainfall in India.
  - Affects Agriculture sector which accounts for 15% of GDP, 58% employment and 70% of dependency for Indian Economy.

  - Affected sectors: Agriculture inputs (seeds, agrochemicals, fertilizers, etc.), Auto, FMCG
  - In April or 1st week of June, monsoon announcement is made by IMD (Indian Meteorological Department) which is a key indicator for the market especially the sectors affected by it directly.
  - Better monsoon $\rightarrow$ Better agriculture output $\rightarrow$ Better GDP $\rightarrow$ Better market performance (but floods are not good)
- **Elections**
  - Political stability impacts market confidence
  - Election Results that are too variable can lead to market volatility
  - Stable governments generally lead to better market performance
  - Policy changes can create sector-specific opportunities
- **Indian Rupee Valuation**
  - Strong rupee benefits importers, weak rupee helps exporters
  - Rising value of rupee leads to better economic health, better fiscal deficit, and lower inflation, good interest rate on FDI (Foreign Direct Investment) Loans, also makes import/export easier and reasonable.
  - Key factors: Trade wars, interest rates, geopolitical tensions, import-export, economic downturn, unforeseen election results, shaky political reforms
  - Sensitive sectors: With rising value of rupee, sectors like Pharma, IT, and Jewelry may face challenges due to reduced competitiveness in exports. While sectors like Oil and Gas, Cement, Tiles, FMCG, Plastic, Chemicals, Tyre, etc. which import raw materials, may benefit from a stronger rupee.

Other economic indicators:

- Forex reserves & trade balance
- Agricultural output & tax structures
- Interest rates & purchasing power parity
- Government infrastructure spending
- Tax structures & fiscal policies
- Foreign Direct Investment (FDI) and FII (Foreign Institutional Investment) trends.
- Per Capita Income & purchasing Power Parity (PPP)
- Employment rates & wage growth

3.2 Industry Analysis

We need to perform industry analysis to get an idea about what industry will be the best to invest in based on its future prospects (more growth potential, high profitability and lower risks). Evaluating sector-specific opportunities and risks:

- Growth potential vs competitive landscape
- Government policies & regulatory environment
- Technological disruption & substitution threats
- Raw material availability & input costs

$$\text{Stock Market growth} \propto \text{Industry potential}$$

Industry potential depends on its growth, risks, opportunity and competition.

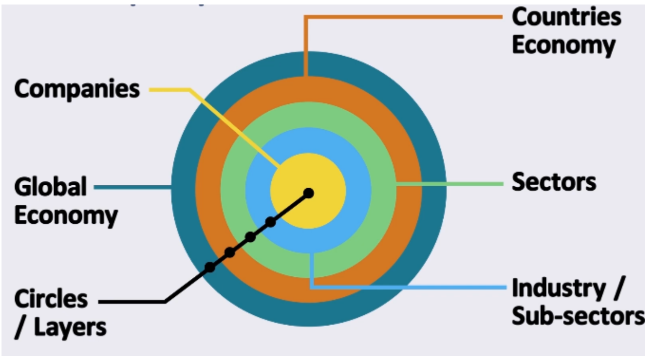

Figure 2: Top-Down Approach in Fundamental Analysis

3.3 Factors affecting Industry

- **Industry Life Cycle Stages**:
  1. Pioneering Stage (Emerging technologies)
  2. Rapid Growth Stage
  3. Maturity & Stabilization
  4. Decline/Diversification
- **Competitive Landscape**:
  - Number of competitors and market share distribution
  - Barriers to entry (capital requirements, regulations)
  - Product differentiation level
- **Government Influence**:
  - Policy support or restrictions
  - Tax incentives/subsidies
  - Environmental regulations
- **Supply Chain Dynamics**:
  - Raw material availability
  - Supplier bargaining power

$$\text{Supplier Power} = \frac{\text{Number of Suppliers}}{\text{Switching Costs}}$$

- **Other Qualitative Factors**:
  - Labour availability and skill levels required
  - Technological advancements and innovation pace
  - Consumer preferences and demand trends. Business cycles and seasonality.
  - Threats from substitutes and new entrants
  - Globalization impact and international competition
- **Quantitative Factors**:
  - Market size and growth rate
  - Profit margins and return on investment (ROI)
  - Turnover and Profitability (Sales) trends
  - Cost Structure (fixed vs variable costs, operating leverage, cost competitiveness within the industry, scalability of operations, and sensitivity to input price changes)
- **Other Factors**:
  - Foreign trade and investments
  - Industrial Production
  - International Events (e.g., discovery of precious metal deposits, new technology breakthroughs, etc.)

4 Company Analysis

Fundamental analysis of individual companies involves both qualitative and quantitative evaluation:

4.1 Qualitative Analysis

(i) Business Model Evaluation

Bussiness model of a company should be simply explainable to even a children. Key questions to assess business viability:

1. What does the company do? (Core operations)
2. Manufacturing capabilities & service delivery mechanisms
3. Plant locations & capacity utilization
4. Raw material sourcing & input costs
5. Client portfolio & end-user demographics
6. Competitive landscape analysis
7. Geographic expansion plans
8. New product pipeline
9. Revenue mix breakdown (revenue model and sources)
10. Best selling Products/Services
11. Distribution channels & sales strategies
12. Regulatory environment constraints
13. Banking partners & audit quality
14. Workforce composition & skill levels
15. Corporate structure (subsidiaries/associates/alliances/parent company)
16. Management type (Professional vs Family-run)

→ SWOT Analysis

SWOT Analysis is a strategic tool used to evaluate the internal and external factors that affect a business. It stands for:

- **Strengths:** Internal attributes that support successful outcomes.
  - Strong management
  - Quality products and services
  - High profit margins
  - Loyal customer base
  - Sound financial position
  - Cost advantage
- **Weaknesses:** Internal limitations or challenges that hinder performance.
  - Lack of funds
  - Product liability risks
  - Absence of innovation
  - Unstable management
  - Competitive intensity
- **Opportunities:** External conditions that the business can capitalize on.
  - Internal and external growth potential

  - Horizontal, vertical, and technical expansion
  - Emerging social trends
  - Favorable macroeconomic conditions
- **Threats:** External elements that could challenge business performance.
  - Litigation risks
  - Changing government policies
  - Intense competition
  - Negative market sentiment
  - Substitutes and pricing pressure

SWOT can be used to evaluate not only businesses, but also industries at large, by identifying the key drivers and potential risks in the sector.

(ii) Competitive Advantage (Economic Moat)

According to Warren Buffett, a company with a sustainable competitive advantage is one that can maintain its profitability and market position over the long term. This is often referred to as having an "economic moat." An economic moat is a unique advantage that protects a company from its competitors, allowing it to maintain higher profit margins and market share. Characteristics of companies with sustainable advantages:

- High customer switching costs
- Strong network effects
- Operational cost leadership (low operation costs)
- Proprietary assets/IP
- Pricing power (near-monopoly status)

Identification Parameters:

1. Substantial cash reserves
2. Consistent performance during economic downturns
3. Brand equity & R&D capability
4. Product line diversification
5. Large-scale operations
6. Regulatory protections (Goverment and regulatory licenses)
7. High customer retention rates

→ Porter's Five Forces (Company Perspective)

Porter's Five Forces is a framework for analyzing the competitive intensity and attractiveness of an industry/company. It helps assess a company's strategic position by examining five key forces:

1. **Rivalry Among Existing Competitors**
   - Number of competitors in the market
   - Market exit barriers
   - Industry size and growth rate
   - Degree of product differentiation or homogeneity
   - Size and loyalty of customer base
   - Threat of horizontal expansion
2. **Threat of New Entrants**

- Capital required to start a new business
- Legal and regulatory barriers to entry
- Access to supplier and distribution networks
- How existing firms react to new entrants

3. **Threat of Substitute Products or Services**
   - Number and availability of substitutes
   - Performance and quality of substitutes
   - Cost incurred by customers when switching to substitutes
4. **Bargaining Power of Buyers**
   - Number of buyers
   - Size and value of individual purchases
   - Switching costs faced by buyers
   - Price sensitivity of buyers
5. **Bargaining Power of Suppliers**
   - Number of suppliers
   - Financial strength of suppliers
   - Cost of switching to alternative suppliers
   - Availability of substitute raw materials or inputs

(iii) Management Analysis

Company's management are the people whom you are trusting with your money and expect that they will generate some returns for you, thus management analysis plays an integral role in your investment's success. Understanding the quality and intent of company management is crucial for investment decisions. Key considerations include:

1. **Promoter Background**
   Review track records for frauds, regulatory issues (SEBI, RBI), and ongoing legal cases.
2. **Remuneration of Senior Management**
   Analyze salary trends, allowances, personal warrent issues, and perks relative to company profits (e.g., year-over-year increase).
3. **Related Party Transactions**
   Scrutinize dealings with promoter-linked entities, joint ventures, interest free loans to relatives, two or more same businesses by same management, or personal loans on behalf of the company.
4. **Management Forecast**
   Evaluate how past projections fared (to which extent the management acieved past projections) and whether recent strategic changes were successfully implemented. Future plans.
5. **Dividend Policy**
   Understand how profits are distributed whether management rewards shareholders or retains earnings. Check the source of money for dividend payments.
6. **Shareholding Patterns and Pledging**
   This analysis should cover the following key areas:
   - Check FII/DII holdings, promoter stake changes, and pledging trends. High pledging implies risk.
   - Who are significant shareholders? Are they increasing or decreasing their stake?
   - Have Mutual Funds and Investment Firms like Insurance companies and Banks invested in the company?
   - Are they increasing or decreasing their stake? (A positive trend shows confidence in the company)

- Note: Professionally-run businesses (e.g., by banks) with transparent governance are generally preferred.
- High promoter pledging or opaque accounting may signal financial instability or value traps.
- But it can also be a sign of confidence if the company is using the funds for growth and expansion (check if company's cash flow is increasing).

7. **Account Manipulation**
   Watch for complex or inconsistent accounting practices that mask the company's real performance.

(iv) Corporate Governance

Corporate governance defines the rules and mechanisms for managing a company in the interest of shareholders. Important elements include:

1. **Ownership and Voting Control**
   Analyze promoter holding, voting rights, and control structures.
2. **Board of Directors Representation**
   Assess board independence and whether it meets current and future needs.
3. **Remuneration Linked to Performance**
   Executive pay should correlate with company performance.
4. **Investor Impact**
   Understand the role and influence of institutional investors.
5. **Shareholder Rights**
   Ensure minority shareholders have adequate protection and influence.
6. **Risk Management**
   Examine how long-term risks are addressed through policies and strategy.

Red Flags: Poor governance undermines transparency, financial reliability, and shareholder trust.

- Are company rules aligned with mission and vision?
- Is every stakeholder served fairly?
- Is the company legally compliant with policies?

Other Qualitative Factors:

- Customer/geographic exposure, market share, disruptive tech
- Regulatory issues, share transactions, management vision
- Reasons for Corporate Issues (e.g., Dividents Payout, Stock Split, Bonus Shares, Rights Issue)
- Corporate culture, employee satisfaction, and retention
- Corporate social responsibility (CSR) initiatives
- Environmental, social, and governance (ESG) factors

4.2 Quantitative Analysis

Quantitative analysis involves evaluating a company's financial health through its financial statements. Financial Modelling is the process of creating a mathematical representation of a company's financial performance, often using historical data to forecast future results.

4.2.1 Interpreting Financial Statements

Types of Financial Statements

- **Standalone Statements:** Represent the financial position of a single entity e.g., subsidiaries. So, if a company has multiple businesses, then a standalone statement of the company will offer details of the financial performance of that particular company.
- **Consolidated Statements:** Represent the financial position of the parent company along with its subsidiaries, associates, or joint ventures. Hence, if you are analyzing the statements of a large organization, then a consolidated statement offers a better picture of the performance of all the companies put together.

Note: Analyzing consolidated financial statements is preferred for in-depth analysis, but they should be reviewed alongside standalone statements. If a company has many subsidiaries, the standalone is too narrow; consolidated gives the full financial picture, and is generally preferred for evaluating the business comprehensively.

Example: When analyzing a company like Tata Power, it's important to look at both standalone and consolidated financial statements. Standalone statements reflect the financials of Tata Power alone, excluding its subsidiaries, while consolidated statements include all group entities under Tata Power, such as its renewable and distribution arms. Therefore, for a complete view of the company's performance and financial health, consolidated statements are more relevant. It's also incorrect to assume that Tata Power's consolidated statements include Tata Steel, as each is a separate listed entity under the Tata Group.

(i) Profit and Loss / Income Statement

- Generally released every quarter (i.e., every 3 months).
- Used to analyze the performance of the company during a specific period, typically a financial year.
- Reflects the company's operating performance in a given year / quarter.

Format Overview

Component	Description and Formula	What to Look for / Analysis
I. Revenue from Operations / Sales	Revenue earned from the firm's core activities (e.g. sales of products/services).	Consistent growth indicates a healthy and scalable core business. Compare YoY and QoQ.
II. Other Income	Non-operational income (e.g. interest, dividend, investment gains, foreign exchange gains, insurance claims, royalties).	High other income may indicate reliance on non-core earnings - check sustainability.

A. Total Revenue	Total Revenue = I + II	Used to assess the complete earning picture. Sudden jumps: investigate source.
III. Cost of Goods Sold / Operating Expenses	Direct costs to produce goods/services (e.g. raw materials, labor).	Rising COGS vs stagnant revenue reduces margins - assess pricing power and input costs.
B. Gross Profit	Gross Profit = A − III	Indicates basic profitability from operations. Monitor gross margin (Gross Profit/Revenue).
IV. Marketing, Advertising, and Promotion Expenses	Costs to promote the business and drive sales.	High marketing costs may reduce profits but can build brand and drive future growth.
V. Sales, General, and Administrative Expenses (SG&A)	Indirect operational expenses: salaries, rent, insurance, office costs, etc. Sometimes also include Depreciation and Amortization	Efficiency in managing these overheads improves operating leverage.
VI. Other Expenses	R&D, foreign exchange loss, impairments, stock-based compensation, utility expenses, CSR expenses, etc.	Volatile or non-recurring items - adjust for normalized earnings. Watch for R&D investment.
C. EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization)	EBITDA = B − (IV + V + VI)	Key measure of core profitability before non-cash and financing impacts. Compare across peers.
VII. Depreciation and Amortization	Non-cash expenses for asset cost allocation. Depreciation: tangible; Amortization: intangible.	High D&A suggests capital-intensive business. Useful for comparing EBITDA vs EBIT.

D. EBIT (Earnings Before Interest and Tax)	EBIT = C − VII	Reflects profitability after accounting for non-cash costs. Good proxy for operating performance.
E. Operating Profit	Operating Profit = D − II	Strips out non-operational income. Reveals true operating health.
VIII. Interest Expenses / Finance Costs	Cost of servicing debt (interest payments).	High interest: assess debt burden. Check interest coverage ratio $\left(\frac{\text{EBIT}}{\text{Interest}}\right)$.
F. PBT (Profit Before Tax)	PBT = E − VIII + II	Profit before taxes. Key for comparing pre-tax margins. Tax holiday or deferred tax can distort comparison.
IX. Income Tax / Provision for Tax	Taxes charged on pre-tax profit.	Effective tax rate = Tax/PBT. Analyze tax planning, deferred tax.
G. PAT (Profit After Tax)	PAT = F − IX + II	Net earnings after tax. Track net profit margin and YoY PAT growth.
H. Net Profit	Net Profit = G	Final earnings figure. Compare with industry average. Adjust for extraordinary items.
I. EPS (Earnings Per Share)	$\text{EPS} = \frac{\text{H}}{\text{Total Outstanding Shares}}$	Key shareholder metric. Use for valuation ratios (P/E). Check for dilution.

(ii) Balance Sheet

- Generally released on the last day of every fiscal quarter (i.e., every 3 months).
- It represents the company's financial position as of a specific date, showing cumulative data since inception—not performance over a period like the income statement.
- Contains detailed info of the company's assets, liabilities, and shareholders' equity.
- **ASSETS = LIABILITIES + SHAREHOLDERS' EQUITY**
- Assets generate revenue → revenue minus costs becomes retained earnings → retained earnings increase equity. Liabilities are either taken to fund assets or meet short-term obligations.

The difference between assets and liabilities is the value attributable to owners (i.e., equity).

- Balance sheets should also be compared with those of other businesses in the same industry since different industries have unique approaches to financing.

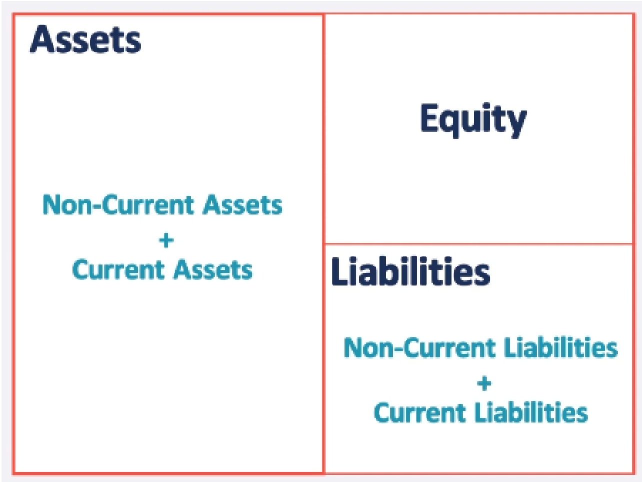

Figure 3: Balance Sheet Structure

Format Overview

Component	Description (with formulas)	What to Look for / Analysis
	EQUITY (Company's actual worth)	
I. Shareholder Equity / Book Value of Company	Book value of equity contributed by shareholders. It is the paid-up capital from primary issuance like IPO, right issues, etc. Note	Indicates how much capital has been injected by shareholders; steady increase indicates investor confidence.
II. Reserve and Surplus / Retained Earnings	Cumulative profits made by the company since inception minus dividends paid.	Consistent growth suggests profitability and internal reinvestment capacity. Declines may indicate high dividend payouts or losses.
III. Non Controlling Assets	Minority interest in a company (Non-controlling ownership in subsidiaries).	Check:- Size: Proportion to total equity. Trend: Growing/shrinking over time. Subsidiary performance: Indicates shared control, profit dilution.

A. Total Shareholder's Fund	$A = \mathrm{I} + \mathrm{II} + \mathrm{III}$ What would be a company's worth if it goes for sale after clearing all its liabilities.	A strong base capital. Should ideally be increasing over time. Useful for calculating Return on Equity (RoE).
LIABILITIES (What company owes)		
NON CURRENT LIABILITIES (Obligations due beyond 1 year)		
III. Long-term Debt / Borrowings	Loans and borrowings with a tenure more than 1 year.	Check for rising debt levels. Used for leverage analysis (Debt-to-Equity). High debt signals financial risk.
IV. Deferred Tax Liabilities	Taxes due for current period but payable in future. Non-cash liability - does not affect immediate cash flow but affects future tax payments.	Indicates timing differences between accounting and tax profits. Stable DTLs are normal.
V. Long-term Provisions	Money set aside for employee benefits, gratuity, etc.	Watch for consistency. Sudden increases might signal rising employee obligations.
VI. Other Long-term Liabilities	Includes deposits like security or rent deposits, deferred credits, etc.	Examine footnotes for breakdown. Evaluate credit terms extended to the firm.
B. Total Long-term Liabilities	$B = \mathrm{III} + \mathrm{IV} + \mathrm{V} + \mathrm{VI}$	Analyze long-term solvency. Should be compared against non-current assets.
CURRENT LIABILITIES (Obligations to be met within 1 year)		
VII. Short-term Debt / Borrowings	Debt with tenure less than 1 year.	High values indicate liquidity risk. Crucial for liquidity ratio analysis (e.g., Current Ratio).
VIII. Trade Payables	Amounts due to suppliers for goods and services.	Increasing payables may improve short-term liquidity but could hurt supplier relationships.
IX. Other Current Liabilities	Unclaimed & Unpaid dividends, overdrafts, current maturities of long-term borrowings.	Understand breakups; higher values may indicate cash crunch.
X. Short-term Provisions	Set aside for short-term obligations like product warranty, employee benefits, gratuity, acceptances (Promises to pay bills (usually trade bills) on due date).	High provisions may be prudent or could hint at underlying risks.

C. Total Current Liabilities	$C = \text{VII} + \text{VIII} + \text{IX} + \text{X}$	Helps assess short-term financial health. Key input in working capital analysis.
D. Total Capital and Liabilities	$D = A + B + C$	Should match total assets. Forms the liability side of the balance sheet.
ASSETS (What company owns)		
NON CURRENT FIXED ASSETS (assets used in the operation of the business and not intended for resale and can't be easily encashed within 1 year)		
XI. Tangible Assets	Physical assets like land, building, equipment.	Useful for calculating asset turnover ratio. Helps determine capital intensity.
XII. Intangible Assets	Non-physical assets: patents (Legal right to exclude others from using an invention), brands (Public perception and identity of a company/product), trademark (Symbol or name identifying a brand, legally protected), goodwill (mentioned only after an acquisition, if purchase price > fair value of net assets due to Intangible asset from reputation, customer loyalty, etc; Goodwill = Purchase Price - (Fair Value of Assets - Liabilities)).	Assess sustainability of competitive advantage. Check amortization schedules.
XIII. Capital Work-in-Progress (CWIP)	Costs incurred on fixed assets under construction.	Large CWIP could indicate expansion or delayed projects.
XIV. Intangible Assets under Development	In-progress intangible developments like R&D, tech platforms.	Examine carefully-may offer future benefits or signify speculative investment.
E. Total Fixed Assets	$E = \text{XI} + \text{XII} + \text{XIII} + \text{XIV}$	Forms backbone of productive capacity. Compare with long-term debt.
NON CURRENT NON FIXED ASSETS (assets not actively used in operations but held for strategic or financial purposes and can't be easily encashed within 1 year)		

XV. Non-current Investments	Investments held for long term (e.g., in subsidiaries, associates).	Strategic holdings. Review for returns, synergy, and stability.
XVI. Long-term Loans and Advances	Loans provided to others with maturity beyond one year.	Must check creditworthiness of borrowers. Could be non-core.
XVII. Other Non-current Assets	Assets not classified above (e.g., deferred (Prepaid costs expensed later example rent, insurance, taxes), deposits).	Inspect for materiality and recurring nature.
F. Total Non-current Assets	$F = E + \mathrm{XV} + \mathrm{XVI} + \mathrm{XVII}$	Total long-term resource allocation. Compare with long-term liabilities.
CURRENT ASSETS (assets expected to be converted into cash, sold, or consumed within one year or the business's operating cycle)		
XVIII. Current Investments	Investments liquid within 1 year (e.g., mutual funds, treasury bills).	High values support liquidity. May earn short-term income.
XIX. Inventories	Items (raw material, work in progress, finished goods) held for resale or production.	Key metric for manufacturing/retail. Watch inventory turnover ratio.
XX. Cash and Cash Equivalents	Includes bank balance, cash on hand, liquid deposits (Bonds).	Key liquidity buffer. Compare with short-term liabilities.
XXI. Short-term Loans and Advances	Loans repayable within a year.	Examine credit quality and collection trends.
XXII. Other Current Assets	e.g., trade receivables (Customer/Vendor payments due for sales), accrued income.	Analyze receivables aging. High values could mean cash is tied up.
G. Total Current Assets	$G = \mathrm{XVIII} + \mathrm{XIX} + \mathrm{XX} + \mathrm{XXI} + \mathrm{XXII}$	Key to working capital analysis. Crucial for short-term financial decisions.
H. Total Assets	$H = F + G$	Should match D (Total Capital and Liabilities). Use to calculate RoA, asset turnover.

Note on Capital Structure:

Authorized capital is the maximum share capital a company is legally allowed to issue, while paid-up capital (also called equity share capital) is the actual amount received from shareholders. Each share has a face value, usually ₹10, which is its nominal worth—not its market price. When a company issues new shares to a new investor, it may do so at a premium if the company's valuation has increased, meaning the investor pays more than the face value. This dilutes existing shareholders' ownership but brings in additional capital. To reward shareholders or improve stock liquidity, companies may issue bonus shares (free additional shares from reserves) or conduct a stock split, where the face value is reduced and the number of shares increases, without changing the total capital. Bonus issues and splits make shares appear more affordable, attracting small investors, though they don't add real value—they're largely psychological and liquidity tools.

(iii) Cash Flow Statement

- Generally released every quarter (i.e., every 3 months).
- Provides insights into cash inflows and outflows from operating, investing, and financing activities.
- Helps assess liquidity, solvency, and financial flexibility.
- One of the most important financial statements, as it shows how well a company generates cash to pay its debt obligations and fund its operating expenses. Also it can not be manipulated easily.

Format Overview

Component	**Description (with formulas)**	**What to Look for / Analysis**
Cash Flow from Operating Activities (from core business operations)		
I. Net Profit Before Tax	Profit before considering taxes and extraordinary items.	Starting point for operating cash flow. Should reconcile with the income statement.
II. Depreciation and Amortization	Non-cash expense for allocation of cost of assets over time.	Added back since it reduces profit but not cash. Higher D&A in asset-heavy firms.
III. Adjustments	Includes non-operating gains/losses (e.g., investment gains), deferred taxes, and stock-based compensation.	Crucial for adjusting accrual-based profit to reflect cash reality. Should be monitored for quality of earnings.
IV. Working Capital Changes	$\Delta WC = \Delta\text{Current Assets} - \Delta\text{Current Liabilities}$	Indicates how much cash is tied up in operations. Large increases in working capital reduce cash flow.

V. Interest Paid	Cash outflows due to interest payments on borrowings.	Deducted under operating cash flows (as per Indian GAAP). Indicates cost of debt.
VI. Tax Paid	Actual tax outflow during the year.	High tax outflow reduces cash; differs from tax expense in P&L due to timing differences.
A. Operating Cash Flow	$A = \mathrm{I} + \mathrm{II} \pm \mathrm{III} \pm \mathrm{IV} - \mathrm{V} - \mathrm{VI}$	Most important segment for cash sustainability. Should be positive and growing over time.
Cash Flow from Investing Activities (from buying/selling assets, investments)		
VII. Net Capital Expenditures	Cash used to acquire or upgrade fixed assets (e.g., plant, machinery).	Usually negative. Shows long-term reinvestment. Excessive capex without returns is a red flag.
VIII. Net Investments	Investments in financial securities or divestments from subsidiaries.	Evaluate whether investments are productive or speculative. Watch for cash inflow from asset sales.
B. Investing Cash Flow	$B = \pm\mathrm{VII} \pm \mathrm{VIII}$	Ideally negative due to capital reinvestment. Sustained positive values may mean asset stripping or low growth.
Cash Flow from Financing Activities (from raising/repaying debt or equity)		
IX. Dividends Paid	Cash paid to shareholders as dividend.	Reflects return to equity holders. High payout may reduce reinvestment ability.
X. Sale or Purchase of Company Stock	Inflows from issuing equity or outflows from buybacks.	Dilution (issue) vs confidence (buybacks). Indicates capital raising or capital return.
XI. Net Borrowings	Cash flows from issuing or repaying loans.	High borrowing inflow could mean cash need; repayment indicates deleveraging.
C. Financing Cash Flow	$C = -\mathrm{IX} \pm \mathrm{X} \pm \mathrm{XI}$	Direction of flow indicates whether company is raising or returning capital.

Net Cash Flow	$= A + B + C$	Should reconcile with change in cash on balance sheet. Positive net cash indicates cash accumulation.

→ **How to Analyze Financial Statements?**

Types of Analysis

1. **Vertical Analysis (Common Size Statements):** Each line item is expressed as a percentage of a base figure, usually total revenue (in income statements) or total assets (in balance sheets).

$$\text{Vertical Analysis} = \left(\frac{\text{Statement Line Item}}{\text{Total Base Figure}}\right) \times 100$$

Example: If Revenue = ₹1,00,000 and COGS = ₹40,000, then:

$$\frac{40,000}{1,00,000} \times 100 = 40\% \quad \text{(COGS as \% of Revenue)}$$

Useful for comparing companies of different sizes and analyzing internal cost structure.

2. **Horizontal Analysis:** Measures the change in a line item over time to detect trends or growth.

$$\text{Horizontal Analysis} = \left(\frac{\text{Current Year Amount} - \text{Base Year Amount}}{\text{Base Year Amount}}\right) \times 100$$

Example: If Revenue in 2022 = ₹1,00,000 and in 2023 = ₹1,20,000, then:

$$\frac{1,20,000 - 1,00,000}{1,00,000} \times 100 = 20\% \quad \text{(Revenue growth)}$$

Useful for trend analysis and assessing year-on-year or quarter-on-quarter performance.

What to Look for while analyzing Financial Statements?

1. **Operating Profit** is the profit a company generates from its core business functions, where interest and tax deduction is excluded

$$\begin{aligned} \text{Operating Profit} = \text{Net Sales} - (&\text{COGS} + \text{Administrative} && (1)\\ &+ \text{Overheads} + \text{Depreciation} + \text{Amortization}) && (2) \end{aligned}$$

2. **Net Profit (Bottom Line)** = Total Revenue - Total Expenses
3. **EBITDA** = Net Profit + Interest + Tax + Depreciation & Amortization
   - Better measure for Capital Intensive Industries (Manufacturing, Oil & Gas, Telecom...)

   EBIT = Net Profit + Interest + Tax
   - Better measure for Service Industries (Technology, Consulting)
4. **Other Income**: Company can try to inflate its net Income by increasing its income from other activities.

5. **Depreciation and Amortization**: Sometimes company changes its accounting policies for calculation of Depreciation & Amortization to inflate its net income.
   Two methods to calculate:
   (i) Straight Line method
   (ii) Accelerated method
6. **Debt/Borrowing**: Lower the debt is better but if company is utilizing the debt the right then its fine. Increasing debt and high interest rates can be very dangerous for the company.
7. **Reserves and Surplus**: It can be good for a company to face financial crisis but too much is not good.
8. **Trade Payables** is the amount payable by a business for goods purchased or services availed as a part of their business operations.
   Trade Receivables is just the opposite i.e., amount receivable.
9. **Cash Flow from Operations** summarizes all cash inflows and outflows from an organization's operational activities.

$$\begin{aligned} \text{CFO} &= \text{PAT} + \text{D\&A} \pm \text{Changes in Current Assets} && (3)\\ &\quad \pm \text{Changes in Current Liabilities} && (4)\\ &= \text{Total Revenue} - \text{Operating Expenses} && (5) \end{aligned}$$

   Note: Positive Cashflow from operations is better. 10-20% less CFO than PAT is ok but else is not so good.
10. **Cash Flow from Investment** - These represent depend on cash inflow and outflow used from investing activities of an organisation.

$$\begin{aligned} \text{CFO} &= \text{Purchase/Sale of Long term Assets} && (6)\\ &\quad + \text{Other Businesses} + \text{Marketable Securities} && (7) \end{aligned}$$

   Note: Negative Cashflow from investments is better.
11. **Cash Flow from Financing Activities** - represents all cash inflow and outflow from financing activities

$$\begin{aligned} \text{CFF} &= \text{Issue/Repurchase (Equity + Debt)} && (8)\\ &\quad - \text{Dividend Payments and other items} && (9) \end{aligned}$$

   Note: Negative Cashflow from financing is better.
12. **Net Cash Flow** is total inflow of cash. (+ve, better) Question any increase/decrease/change in important line items and why it changes from previous year/quarter.

→ How to Read Annual Reports of a Company?

Annual Report is a financial document (having both qualitative and quantitative information) published annually to showcase a company's business operations and financial condition for the concerned fiscal year. It is the most trusted source for getting any data related to a company since it is a public document and verified by the the National Financial Reporting Authority (NFRA). While building any financial model when you fetch data from websites like moneycontrol, you should verify atleast the broad line items from the annual report.

Key Sections to Focus On

1. **Starting Pages** → Vision, branding (should not be too much)
2. **Product's Suite**

3. **Key Performance Indicators**
4. **Compliances**
5. **Chairman's Letter**: Its objective is to state the reasons regarding how a company addressed its failures and related issues. Provides in-depth analysis of the overall business performance. Indicates challenges and risks faced by company. Conclude with financial plans and business direction for upcoming years.
6. **MD's Message**
7. **Risk Mitigation**
8. **Board of Directors**
9. **Business Profile**: Revenue generated, details about products/services manufactured/provided, subsidiaries owned, capital market in which it competed, risk factors, company policy, new products/services, etc.
10. **Financial Statements**
11. **Notes to Financial Statements**
12. **Management Discussion and Analysis**: See management is answering their questions, avoiding or correcting it?
13. **Industry Insight**
14. **Share Holding Pattern**
15. **Debt Analysis**
16. **Board Meetings**
17. **Salary of Management**
18. **Related Party Transactions**
19. **Corporate Governance**

→ How to Read and Understand Quarterly Results?

Quarterly Reports are summary of unaudited financial statements issued by company every quarter (3 months i.e., 4 times a year - June, Sept, Dec, March)

Quarterly Reports analysis

- Helps to access the performance of the company invested in or to make informed decisions
- Consist of Operational Highlights, Press release, FAQ sheet, Quarterly Presentations, Shareholding Pattern, Earning Call Transcript, Management Commentary
- It can show the seasonality or trend of the company and industry. You should find reasons for the same
- Compare the Quarterly Results to previous one and to the previous year's same quarter
- If you find some manipulations in the results you should see the inside story and treat it like a red flag ▷

→ Red Flag Analysis of a Company?

Identifying red flags is crucial in avoiding fundamentally weak or potentially fraudulent companies. Watch out for the following warning signs:

1. **Declining Revenues**
   Consistent fall in top-line growth indicates weak demand or operational issues.
2. **Rising Debt-to-Equity Ratio**
   Excessive leverage raises financial risk and reduces solvency.
3. **Volatile Cash Flows**

Unpredictable cash inflows/outflows reflect operational instability.

4. **Stock Price Fluctuation**
   Extreme or unexplained volatility in share price.
5. **Pending Legal Cases**
   Ongoing litigation could result in financial or reputational damage.
6. **Falling Promoter Holding**
   Continuous decrease signals lack of confidence in the business.
7. **High Promoter Pledging**
   Rising pledging of shares is a risk to ownership stability.
8. **High NPAs (especially for Banks)**
   Indicates poor lending practices and financial deterioration.
9. **Delayed Reports**
   Late filing of annual or quarterly reports may suggest governance issues.
10. **Over-Optimistic Financials**
    Results that appear too good to be true often are.
11. **Auditor Red Flags**
    Any mention of "misstatements" or exceptions in audit reports. Read the auditor's report ("Summary of misstatements") carefully.
12. **Unusual Accounting Practices**
    Frequent changes in accounting methods or policies.
13. **Changes in Financial Reporting Style**
    May indicate manipulation or hiding of actual performance.
14. **Sudden Management Changes**
    Frequent turnover raises stability concerns.
15. **Last-Minute Bulk Sales**
    Sales booked mostly in the last few days of a period are suspicious.
16. **Asset Revaluation Surprises**
    Sudden large changes in fixed/intangible asset value.
17. **Complex Transactions**
    Unnecessarily complicated deals may hide risks or fraud.
18. **Performance-Linked Bonuses**
    Incentives tied to manipulated metrics can misalign interests.
19. **Strange behaviour of key Ratios**
    This analysis should cover the following key areas:
    - Gross Profit Margin ↑ but Sales ↓
    - Return on Equity (ROE) ↓ but Price-to-Book (P/B) Ratio ↑
20. **Rising Debt or Inventory**
    Build-up in debt or unsold goods implies weak demand or inefficiency.

Note: Multiple red flags together strengthen the case for caution or avoidance. Always investigate the underlying causes.

4.2.2 Ratio Analysis

Ratio analysis is a quantitative method of evaluating a company's financial performance by comparing various financial metrics. It helps investors and analysts assess profitability, liquidity, solvency, and efficiency.

(i) Solvency / Financial Risk Ratios

Used to measure an entrise's ability to meet its long term debt obligations.

Ratio Name	Formula	Analysis & Interpretation
Debt to Equity Ratio	$\frac{\text{Total Debt}}{\text{Total Equity}}$	• Measures financial leverage • How much business is funded by debt compared to equity distribution • Ideal: < 1 • Compare with industry peers (e.g., acceptable > 2 for capital-intensive industries like Power, Banks)
Debt Ratio	$\frac{\text{Total Debt}}{\text{Total Assets}}$	• Shows %age of assets financed by debt • Safe threshold: < 0.5 • > 0.7 indicates aggressive leverage and high risk • Critical for cyclical industries • Generally hight debt ratio in industries like Power, Banks but with stable cash flows, yet should not be > 0.75
Debt to Capital Ratio	$\frac{\text{Total Debt}}{\text{Debt + Shareholders' Equity}}$	• shows the debt company has compared to its equity • Similar analysis to Debt Ratio • Ideal: < 0.5
Interest Coverage Ratio (ICR)	$\frac{\text{EBIT}}{\text{Interest Expense}}$	• Debt servicing capacity • How many times a company can cover its current interest payments with its available earnings • Safe: > 3 for most industries, < 1.5 indicates distress risk • For utilities like Power companies: > 2 and < 3 is recommended
Debt Service Coverage Ratio (DSCR)	$\frac{\text{Operating Profit}}{\text{Total Debt Service}}$	• Cash flow adequacy for debt payments • Healthy: $1.5 < DSR < 2$ • Debt Service = interest + principal repayments
Financial Leverage Ratio	$\frac{\text{Total Assets}}{\text{Shareholders' Equity}}$	• How much the assets belongs to shareholders than debt creditors • Higher ratio = More debt reliance (Lower better)

Proprietary Ratio (Equity Ratio)	$\frac{\text{Shareholders' Funds}}{\text{Total Assets}}$	• To what extent Shareholders' funds have been invested in the assets of the business • Ideal: > 0.5; < 0.3 indicates high creditor stake

(ii) Liquidity Ratios

Used to determine a debtor's ability to pay off current debt obligations without raising external capital.

Ratio Name	Formula	Analysis & Interpretation
Current Ratio	$\frac{\text{Current Assets}}{\text{Current Liabilities}}$	• Short-term debt paying ability • Ideal: 1.33 - 3 • > 3: Inefficient asset utilization, funds are either idle or locked up in inventories and receivables • < 1: Potential liquidity crisis
Quick Ratio (Acid Test)	$\frac{\text{Current Assets}-\text{Inventory}}{\text{Current Liabilities}}$	• Strict liquidity measurement • Ideal: $1 <$ Cash Ratio < 2.5 • > 2.5 suggests improper utilization of short term funds • Critical for industries with slow-moving inventory
Cash Ratio	$\frac{\text{Cash} + \text{Cash Equivalents}}{\text{Current Liabilities}}$	• How quick and to what extent, company can repay its current dues with help of its readily available assets • Ideal: 0.5 - 1 • > 1: Excessive idle cash • < 0.2: High default risk

(iii) Efficiency Ratios

Measures company's ability to use its assets and manage its liabilities effectively in the current period or in short term.

Ratio Name	Formula	Analysis & Interpretation

Inventory Turnover Ratio	• $\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$ • Inventory Turnover Days $= \frac{365}{\text{Inventory Turnover Ratio}}$	• Measures inventory management efficiency • Higher = Better (industry dependent), Manufacturing: 6-8x ideal, Retail: 12-15x desirable • Inventory Turnover Days tells on average how many days the company's inventory is held until it is sold. Generally Lower better, but much low means company is not fulfilling its demand.
Receivable Turnover Ratio	• $\frac{\text{Net Credit Sales}}{\text{Average Account Receivables}}$ • Account Receivable Days $= \frac{365}{\text{Receivable Turnover Ratio}}$	• How quick company collects bills from its customers, Higher = Faster conversion • Net Credit Sales = Sales on credit - Sales returns - Sales Allowances • Account Receivable Days is how many days does the company takes to collect cash from its credit sales from the customers
Payable Turnover Ratio	• $\frac{\text{COGS}}{\text{Average Payables}}$ • Payable Days $= \frac{365}{\text{Payable Turnover Ratio}}$	• Payment discipline measurement • Lower = Better credit terms, but high canmean the company has favourable credit terms • Payable Days is the average payment period
Total Asset Turnover Ratio	$\frac{\text{Net Sales}}{\text{Average Total Assets}}$	• Ability of company to generate sales from its assets • Higher = Better capital use • 0.5-2x = Common range, For Capital-intensives: 0.3-0.8x normal

Working Capital Turnover	• $\frac{\text{Net Sales}}{\text{Average Working Capital}}$ • Working Capital = Current (Assets - Liabilities) • Working Capital Cycle = (Inventory + Receivable - Payable) Days	• Reflects the amount of operating profit needed to maintain a given level of sales • Higher = Better (5-10x ideal) • Working Capital is the amount of cash business can safely spending • Working Capital / Cash Conversion Cycle is the time it takes to convert total net working capital into cash. Ideally should be negative

(iv) Financial / Profitability Risk Ratios

Growth Ratios

Indicates how fast the business is growing, Higher better.

Ratio Name	Formula
Operating Profit Margin	$\frac{\Delta\text{Operating Profit}}{\text{Net Sales}} \times 100$
Gross Profit Margin	$\frac{\Delta\text{Gross Profit}}{\text{Net Sales}} \times 100$
EPS Growth	$\frac{\Delta\text{EPS}}{\text{Prior EPS}} \times 100$

Similarly, we can calculate EBITDA margin, PAT margin, etc.

We can also calculate their Compounded Annual Growth Rate (CAGR) or simple percentage change as follows, where n is the number of periods (e.g., years):

$$\text{CAGR} = \left(\frac{\text{End Value}}{\text{Start Value}}\right)^{\frac{1}{n}} - 1$$

$$\%\text{ change} = \frac{\text{Final value} - \text{Initial Value}}{\text{Initial Value}} \times 100$$

Return Ratios

Return on Investment (ROI)	$\frac{\text{Net Return}}{\text{Cost}} \times 100$	• It is just a basic return measurement for any investment you make.

Return on Invested Capital (ROIC)	$\frac{\text{EBIT}(1-\text{Tax})}{\text{Debt}+\text{Equity}}$	• How well a company is using its money to generate returns • ROIC > WACC = Value creation • If the company uses debt to finance investments, calculate WACC to determine if ROIC exceeds the cost of capital (including the interest rate on debt), indicating value creation.
Return on Equity (ROE)	$\frac{\text{PAT}}{\text{Total Equity}} \times 100$	• How well is the firm in generating returns on the investment received from its shareholders • Higher Better • High ROE with high debt = Risk, Companies can inflate ROE by increasing it's debt.
Return on Capital Employed (ROCE)	$\frac{\text{EBIT}}{\text{Capital Employed}}$	• Overall fficiency to generate profits from capital • Capital Employed = Equity + Debt (Long Term + Short Term) • Higher Better, ROCE > WACC

Return on Assets (ROA)	$\frac{\text{EBIT or PAT}}{\text{Total Assets}}$	• Measures how efficiently a company uses its assets to generate earnings
		• Higher ROA is better
		• ROA can be increased by:
		  – Increasing EBIT (better pricing, economies of scale, cost efficiency)
		  – Decreasing assets (reducing receivables, better inventory turnover, higher asset utilization, divesting low-margin businesses, leasing instead of buying)

Weighted Average Cost of Capital (WACC)	$\frac{\text{Equity}}{\text{Total Capital}} \times$ Cost of Equity OR Expected rate of Return+ $\frac{\text{Debt}}{\text{Total Capital}} \times \text{Interest Rate} \times (1 - \text{Tax Rate})$	• Represents a firm's blended cost of capital (equity + debt) • Required return for a capital budgeting project • Lower is better for investors; higher implies a company must generate more returns

Cash Flow Ratios

Cash Flow Per Share	$\frac{\text{Operating CF}-\text{Pref. Div}}{\text{Outstanding Shares}}$	• Real cash generation, it is the net cash a company produces on a per share basis • Compare with EPS • CFPS > EPS = Quality earnings
Free Cash Flow Per Share (FCFPS)	$\frac{\text{Free Cash Flow}}{\text{Total Outstanding Shares}} = \frac{\text{Operating CF}-\text{CAPEX}}{\text{Outstanding Shares}}$	• Indicates a company's ability to pay debt, dividends, buy back stock, and grow the business • Higher FCFPS is better • Increasing FCF implies potential increase in share price • FCF = cash left after operating expenses and asset-related capital expenditure • Capital Expenditure (CAPEX) = funds used to acquire or upgrade physical assets like plant, property, and equipment
Free Cash Flow to Equity (FCFE)	PAT + Depr. & Amor. − CAPEX ± Change in current assets ± Change in current liabilities	• Cash available to equity holders after meeting all debt obligations
Free Cash Flow to Company (FCFF)	EBIT(1 − Tax) + D&A − CAPEX ± Change in current assets ± Change in current liabilities	• Cash flow available to equity and debt holders in the company
Dividend Yield	$\frac{\text{Annual Div/Share}}{\text{Stock Price}} \times 100$	• measures how much a firm pays out in dividents each year relative to stock price • 2-4% = Healthy; >6% = Sustainability check

→ Dupont Analysis of ROE

The DuPont analysis is a financial framework that decomposes Return on Equity (ROE) into three key components to provide insights into a company's profitability, operational efficiency, and financial structure. By breaking down ROE, it helps investors and analysts understand whether high returns are driven by strong profits, efficient asset use, or significant debt, making it a valuable tool for fundamental analysis.

Formula

The DuPont analysis expresses ROE as the product of three ratios:

$$\text{ROE} = \left(\frac{\text{Net Income}}{\text{Sales}}\right) \times \left(\frac{\text{Sales}}{\text{Assets}}\right) \times \left(\frac{\text{Assets}}{\text{Shareholders' Equity}}\right)$$

Where:

- **Profit Margin** ($\frac{\text{Net Income}}{\text{Sales}}$): Measures how much profit a company generates per dollar of sales, reflecting its ability to control costs and pricing power.
- **Asset Turnover** ($\frac{\text{Sales}}{\text{Assets}}$): Indicates how efficiently a company uses its assets to generate sales, showing operational efficiency.
- **Financial Leverage** ($\frac{\text{Assets}}{\text{Shareholders' Equity}}$): Represents the degree to which a company uses debt to finance its assets, amplifying returns but increasing risk.

This decomposition allows for a detailed analysis of the factors contributing to a company's ROE, facilitating comparisons across firms and industries.

Components of DuPont Analysis

- **Profit Margin**: A high profit margin indicates strong profitability, often due to premium pricing, cost efficiency, or a favorable product mix. For example, technology companies like Apple often have high margins due to brand strength.
- **Asset Turnover**: A higher ratio suggests efficient use of assets to generate revenue. Retail companies typically have high asset turnover due to rapid inventory turnover, while capital-intensive firms may have lower ratios.
- **Financial Leverage**: A high leverage ratio indicates reliance on debt, which can boost ROE but also increases financial risk. Companies with stable cash flows can sustain higher leverage safely.

Steps in DuPont Analysis

1. **Collect Financial Data**: Obtain Net Income, Sales (Revenue), Total Assets, and Shareholders' Equity from the company's financial statements (income statement and balance sheet).
2. **Calculate Profit Margin**: Divide Net Income by Sales to determine profitability per dollar of revenue.
3. **Calculate Asset Turnover**: Divide Sales by Total Assets to assess asset efficiency.
4. **Calculate Financial Leverage**: Divide Total Assets by Shareholders' Equity to measure debt usage.
5. **Compute ROE**: Multiply the three ratios to obtain ROE and verify by calculating ROE directly ($\frac{\text{Net Income}}{\text{Shareholders' Equity}}$).
6. **Analyze Results**: Interpret each component to understand the drivers of ROE and compare with industry peers.

Example: Apple Inc. (Fiscal Year Ended September 28, 2024)

To illustrate the DuPont analysis, consider Apple Inc.'s financial data for the fiscal year ended September 28, 2024, sourced from its annual report:

Using these figures, we calculate each component of the DuPont formula:

1. **Profit Margin**:

$$\text{Profit Margin} = \frac{\text{Net Income}}{\text{Sales}} = \frac{93,736}{391,035} \approx 0.2397 \text{ or } 23.97\%$$

Metric	Value (in millions)
Net Income	\$93,736
Total Revenue (Sales)	\$391,035
Total Assets	\$364,980
Shareholders' Equity	\$56,950

Table 10: Financial Data for Apple Inc., Fiscal Year 2024

Apple generates approximately 24 cents in profit for every dollar of sales, reflecting strong profitability due to its premium pricing and brand strength.

2. **Asset Turnover**:

$$\text{Asset Turnover} = \frac{\text{Sales}}{\text{Assets}} = \frac{391,035}{364,980} \approx 1.0714$$

Apple generates \$1.07 in sales per dollar of assets, indicating moderate efficiency in asset utilization.

3. **Financial Leverage**:

$$\text{Financial Leverage} = \frac{\text{Assets}}{\text{Shareholders' Equity}} = \frac{364,980}{56,950} \approx 6.4081$$

For every dollar of equity, Apple has \$6.41 in assets, suggesting significant use of debt or liabilities, though mitigated by substantial cash reserves.

Combining these, the ROE is:

$$\text{ROE} = 0.2397 \times 1.0714 \times 6.4081 \approx 1.6458 \text{ or } 164.58\%$$

To verify, calculate ROE directly:

$$\text{ROE} = \frac{\text{Net Income}}{\text{Shareholders' Equity}} = \frac{93,736}{56,950} \approx 1.6458 \text{ or } 164.58\%$$

The calculations align, confirming the accuracy of the DuPont analysis.

Interpretation

The DuPont analysis reveals that Apple's exceptionally high ROE of 164.58- **High Profit Margin (23.97%)**: Apple's ability to command premium prices for its products (e.g., iPhones, MacBooks) and maintain cost efficiency results in strong profitability. - **Moderate Asset Turnover (1.0714)**: Apple uses its assets efficiently to generate sales, though not as aggressively as retail or manufacturing firms with higher turnover ratios. - **High Financial Leverage (6.4081)**: Apple's low shareholders' equity relative to assets, partly due to share buybacks, amplifies ROE. However, its large cash reserves (e.g., \$29.97 billion in cash as of September 2024) reduce the risk associated with high leverage.

This analysis highlights Apple's strong profitability and strategic use of leverage, tempered by moderate asset efficiency. Investors can compare these components to industry peers to assess relative performance.

(v) Valuation Ratios

Helps to develop a sense of how the stock price is valued by market participants

Ratio Name	**Formula**	**Analysis & Interpretation**
Earning Per Share (EPS)	$\frac{\text{Net Income - Preffered Dividents}}{\text{Outstanding Shares}}$	• To assess a company's profitability on a per-share basis • A rising EPS over time signals improving profitability, a positive sign for investors.
Price-to-Earnings (P/E)	$\frac{\text{Market Price}}{\text{EPS}}$	• How much the market is willing to pay for every rupee of company's current earnings • Sector averages vary widely, Compare with peers, industry average and historical P/E of the company. • Low P/E = stock might be undervalued, low or negative growth expectation • High P/E = Growth expectations, stock might be overvalued
Price-to-Book (P/B)	• $\frac{\text{Market Price}}{\text{Book Value/Share}}$ • Book value = Total assets - Intangible assets - Liabilities • Book value per share = $\frac{\text{Book value}}{\text{Outstanding Shares}}$	• How much shareholders are paying for a company's net assets • Book value per share indicates a firm's net asset value on per share basis • High P/B ratio indicate market is ready to pay premium on book value because of future growth of company or value of intangible assets • More useful for asset heavy industries (and not for technology and service industries)

PEG Ratio	$\frac{\text{P/E}}{\text{EPS Growth Rate}}$	• Denotes rupee amount a shareholder needs to pay for a stock of a company for earning ₹1 of its income • <1 = Undervalued • >1.5 = Overvalued • not a goog metric for asset heavy businesses
Price-to-Cash Flow	$\frac{\text{Market Price}}{\text{Operating Cash Flow/Share}}$	• Rupee value an investor is willing to pay for the cash flow generated • Compare with P/E ratio, Lower better • if ther is a gap between net profit and operating cash flow, the company should be investing money in expansion and not in renewal
EV/EBITDA	$\frac{\text{Enterprise Value}}{\text{EBITDA}}$	• Capital structure neutral, used to compare the entire value of a business with the amount of EBITDA it earns on an annual basis • 8-12x = Fair value range • <6x = Acquisition target
Price-to-Sales (P/S)	$\frac{\text{Market Cap}}{\text{Revenue}}$	• Shows how much themarket values every rupee of compan's sales, Higher better • Compare with peers • Best ratio to watch if company is at temporary losses.

(vi) Ratios For Banks

Ratio	Formula	Analysis
CASA (Current Account Saving Account) Ratio	$\frac{\text{CASA Deposits}}{\text{Total Deposits}}$	• Indicates how much of total bank deposits are in current and savings accounts • Higher is better (lower cost of funds), But not always, balance needed • Typically, banks offer no interest on current a/c, 2–3% on savings a/c

Credit to Deposits % (LTD)	$\frac{\text{Bank's Total Loan Disbursed}}{\text{Total Deposits Received}}$	• Indicates how much of deposits are lent out • Too low: bank not earning much • Too high: liquidity risk in unforeseen events • Ideal between 80% - 90%
Interest Expended to Total Funds %	$\frac{\text{Interest Expense}}{\text{Total Deposits}}$	• Shows what portion of income is spent on interest for borrowed money • Lower is better • <10% implies sufficient liquidity • >10% is generally not good
Interest Income to Total Funds %	$\frac{\text{Net Interest Income}}{\text{Total Deposits in Bank}}$	• Measures bank's earning from interest relative to total funds • Net Interest Income = Interest Earned - Interest Paid • Higher is better
Advances Growth %	$\frac{\text{Last Year Adv.} - \text{This Year Adv.}}{\text{Last Year Advances}} \times 100$	• Shows how much bank advances have grown year-on-year • Advances = loans given to individuals/businesses to meet working capital needs • Higher is better
Non-Performing Assets (NPA)	GNPA% = $\frac{\text{Gross NPA}}{\text{Total Advances}} \times 100$ NNPA% = $\frac{\text{Net NPA}}{\text{Total Advances}} \times 100$	• Loans overdue by more than 90 days are NPAs • Lower is better, shows better loan book quality • Gross NPA is the total value of advances that have become NPA in a particular quarter/financial year • Net NPA = GNPA - Provisions (actual bad loans after accounting provisions) • Provisions are the funds set aside by Company/Bank as assets to pay for anticipated future losses/taxes.
Cost to Income Ratio %	$\frac{\text{Operating Expenses}}{\text{Operating Income}}$	• Measures efficiency: cost incurred to earn income • Lower is better

Operating Cost to Assets Ratio %	$\frac{\text{Operating Expenses}}{\text{Average Assets}}$	• Measures operating efficiency of bank relative to asset base • Lower is better - more profitable and efficiently managed
Cost of Liabilities %	$\frac{\text{Total Interest on Borrowings and Deposits}}{\text{Total Borrowings and Deposits}}$	• Indicates rate paid on current debt • Lower is better — improves margins and profitability
Capital Adequacy Ratio (CAR)	$\frac{\text{Total Capital}}{\text{Risk Weighted Assets}}$	• Shows bank's ability to absorb potential losses • Higher is better - protects depositor money and ensures stability
Net Interest Margin (NIM) Ratio	$\frac{\text{Interest Earned} - \text{Interest Paid}}{\text{Assets that Bank Holds}}$	• Measures profitability from lending/investment activities • Must be positive; higher is better
Yield on Advances %	$\frac{\text{Total Interest Earned}}{\text{Average Loans \& Advances}}$	• Measures return bank earns on its loans • Higher is better

(vii) Market Meter Ratios

Ratio Name	**Formula**	**Analysis & Interpretation**

Market-Cap to GDP (Buffett Indicator)	$\frac{\text{Total Market Cap}}{\text{GDP of country}} \times 100$	• Broad market valuation
		• For India:
		  – $> 75\%$: Overvalued (Sell signal)
		  – $< 75\%$: Undervalued (Buy signal)
		• US Historical Avg: 100-120%

Debt to GDP Ratio	$\frac{\text{Total Government Debt}}{\text{GDP of country}} \times 100$	• Sovereign risk measure (country's ability to repay its debt) • By Debt here we mean, the fiscal deficit (the difference between the total income of government and its total expenditure) • India Target: $< 64\%$ • $> 64\%$: can lead to Market crash • Developed Nations: $80 - 120\%$ common

Nifty/Sensex P/E Ratio	$\frac{\text{Total Market Cap of Nifty 50}}{\text{Total Earnings of Nifty 50}}$	• Market valuation gauge • Compare with 10-Yr Average • India Historical Range: 18-28 P/E • >25: Caution warranted

4.2.3 Valuation Analysis

Intrinsic Value

Intrinsic Value is the anticipated or calculated value of a stock, currency, or product based on fundamentals rather than market price.

Key Points to Remember

1. It's all about future cash flows.
2. You must discount the cash flows to the present day.
3. Understand the business model clearly.
4. It gives a rough figure not an exact number.
5. Always begin with assumptions.

(i) Method 1: Asset-Based Valuation

Formula:

$$\text{Value per share}$$
$$= \frac{\text{Market Capitalization of Company} + \sum(\text{Percentage Holding in company} \times \text{Market Cap of company})}{\text{Total Number of Equity Shares in Parent / Holding Company}}$$
$$= \frac{\text{Net Assets} - \text{Preference Shares Capital} - \text{Debt}}{\text{Number of Equity Shares}}$$

Use Case: Primarily for holding/parent companies with multiple assets. **Note:** Suggests buying when there's a *discount rate* (e.g., 25%).

(ii) Method 2: Market Based Valuation

- **Expected rate of Dividents:**

$$\frac{\text{Profit available for Dividents}}{\text{Paid-up Equity Capital}} \times 100$$

- **Expected rate of Earnings:**

$$\frac{\text{Profit After Tax}}{\text{Paid-up Equity Capital}} \times 100$$

- **Value per Share:**

$$\frac{\text{Expected rate of Earnings or Dividents}}{\text{Normal Rate of Return}} \times \text{Equity paid up Capital}$$

(iii) Method 3: Comparable Company Analysis (Relative Valuation)

- Aims to derive a stock's theoretical price using the price multiples of similar companies
- It involves the use of valuation ratios like P/E ratio, P/B Ratio, EV/EBITDA, EV/Sales, etc.
- Compare company with its historical and Industrial Averages and with its peers.

(iv) Method 4: Dividend Discount Model (DDM)

Use Case: Companies that pay stable dividends.

Formula (Gordon Growth Model):

$$\text{Fair Value} = \frac{D_1}{r - g}$$

Where:

- D_1 = Expected dividend next year
- r = Cost of equity
- g = Dividend growth rate

(v) Method 5: Discounted Cash Flow (DCF)

The Discounted Cash Flow (DCF) method is a valuation technique used to estimate the intrinsic value of a company, project, or asset based on its expected future cash flows. It incorporates the time value of money, recognizing that a dollar today is worth more than a dollar in the future due to its potential earning capacity. The DCF method is widely used in investment finance, corporate financial management, and for valuing companies, particularly for long-term investment decisions [**?**].

Formula

The DCF value is calculated as the sum of the present values of future cash flows over a forecast period, often combined with a terminal value to account for cash flows beyond that period:

$$\text{DCF} = \sum_{t=1}^{n} \frac{\text{CF}_t}{(1+r)^t} + \frac{\text{TV}}{(1+r)^n}$$

Where:

- CF_t: Cash flow in year t,
- r: Discount rate,
- n: Number of forecast periods,
- TV: Terminal value, representing cash flows beyond the forecast period.

Components of DCF

- **Cash Flows ($\mathbf{CF}_t$)**: These are typically free cash flows to the firm (FCFF), which represent cash available to all investors (debt and equity holders) after operating expenses, taxes, and capital expenditures. Alternatively, free cash flows to equity (FCFE) can be used for equity valuation, reflecting cash available to shareholders after debt payments. FCFF is preferred for enterprise valuation, as it captures the total value of the company.

- **Discount Rate (r)**: The rate used to discount future cash flows to their present value, reflecting the investment's risk. For FCFF, the weighted average cost of capital (WACC) is commonly used, calculated as the weighted average of the cost of equity and cost of debt. For FCFE, the cost of equity is used. The WACC accounts for the company's capital structure and risk profile.
- **Terminal Value (TV)**: Since companies are assumed to generate cash flows indefinitely, the terminal value estimates the present value of all cash flows beyond the forecast period. It is often calculated using the perpetuity growth model:

$$\text{TV} = \frac{\text{FCFF}_{n+1}}{r - g}$$

  Where FCFF_{n+1} is the cash flow in the year after the forecast period, and g is the perpetual growth rate (typically 2–3%, reflecting long-term economic growth).

Steps in DCF Valuation

1. **Forecast Free Cash Flows**: Estimate FCFF or FCFE for a period (e.g., 5–10 years) based on financial statements, revenue growth, operating margins, and capital expenditures.
2. **Determine the Discount Rate**: Calculate the WACC using the cost of equity (e.g., via the Capital Asset Pricing Model) and cost of debt, weighted by the company's capital structure.
3. **Calculate Present Value of Cash Flows**: Discount each year's cash flow to the present using the formula $\frac{\text{CF}_t}{(1+r)^t}$.
4. **Estimate Terminal Value**: Use the perpetuity growth model or an exit multiple (e.g., based on EBITDA) to estimate cash flows beyond the forecast period.
5. **Discount Terminal Value**: Discount the terminal value to the present using $\frac{\text{TV}}{(1+r)^n}$.
6. **Sum Present Values**: Add the discounted cash flows and terminal value to get the enterprise value.
7. **Adjust for Equity Value**: Subtract net debt (total debt minus cash) from the enterprise value to obtain the equity value. Divide by the number of shares outstanding to get the per-share value.

Example

Consider a company with the following projected FCFF over five years, a WACC of 5%, and an initial investment of $11 million (e.g., for acquiring the company):

Year	Cash Flow	Discounted Cash Flow
1	$1,000,000	$952,381
2	$1,000,000	$907,029
3	$4,000,000	$3,455,350
4	$4,000,000	$3,290,810
5	$6,000,000	$4,701,157

Table 14: Discounted Cash Flows for the Company

The present value of each cash flow is calculated as:

$$\text{PV of CF}_t = \frac{\text{CF}_t}{(1+r)^t}$$

For each year:

- Year 1: $\frac{1,000,000}{(1+0.05)^1} = 952,381$
- Year 2: $\frac{1,000,000}{(1+0.05)^2} = 907,029$
- Year 3: $\frac{4,000,000}{(1+0.05)^3} = 3,455,350$
- Year 4: $\frac{4,000,000}{(1+0.05)^4} = 3,290,810$
- Year 5: $\frac{6,000,000}{(1+0.05)^5} = 4,701,157$

Summing these gives the total DCF for the forecast period:

$$\text{Total DCF} = 952,381 + 907,029 + 3,455,350 + 3,290,810 + 4,701,157 = 13,306,727$$

To account for cash flows beyond year 5, assume a perpetual growth rate (g) of 2%. The FCFF in year 6 is estimated as $\text{FCFF}_6 = \text{FCFF}_5 \times (1+g) = 6,000,000 \times 1.02 = 6,120,000$. The terminal value is:

$$\text{TV} = \frac{6,120,000}{0.05 - 0.02} = \frac{6,120,000}{0.03} = 204,000,000$$

Discount the terminal value to the present (year 0):

$$\text{PV of TV} = \frac{204,000,000}{(1+0.05)^5} = \frac{204,000,000}{1.2762815625} \approx 159,863,013$$

The enterprise value is:

$$\text{Enterprise Value} = 13,306,727 + 159,863,013 = 173,169,740$$

If the company has $2 million in net debt, the equity value is:

$$\text{Equity Value} = 173,169,740 - 2,000,000 = 171,169,740$$

Assuming 1 million shares outstanding, the fair price per share is:

$$\text{Price per Share} = \frac{171,169,740}{1,000,000} \approx 171.17$$

The Net Present Value (NPV) for the investment decision is:

$$\text{NPV} = 173,169,740 - 11,000,000 = 162,169,740$$

Since the NPV is positive, the investment is worthwhile.

4.2.4 Other Analysis

(i) Business Contribution Analysis

When analyzing a company's financials, it's crucial to assess its revenue distribution across different geographies. A higher proportion of revenue from the domestic market, such as India, indicates reliance on local economic conditions, regulatory frameworks, and currency stability. Conversely, significant revenue from international markets exposes the company to foreign exchange fluctuations, geopolitical risks, and global economic trends.

Example:

Region	**Revenue**	**Percentage**
India	88,087	22.55%
Rest of World	233,861	77.45%
Total	**301,938.4**	**100.00%**

Table 15: Business Contribution by Region

Business Contribution Analysis: The company derives approximately **22.55%** of its revenue from India and a significant **77.45%** from the Rest of the World. This indicates a strong global presence with limited domestic reliance. While this diversification reduces dependency on a single market, it also exposes the company to foreign exchange risks and global economic fluctuations. For investors seeking exposure to companies with strong domestic demand, this company may not be ideal. However, if the goal is to invest in a company benefitting from global diversification, this company shows significant global footprint.

(ii) Shareholding Analysis

The shareholding pattern of a company reveals its ownership structure, offering insights into governance, financial health, and market perception. Key aspects include promoter holdings, pledging, and stakes of Foreign Institutional Investors (FIIs) and Domestic Institutional Investors (DIIs).

Promoter Shareholding and Pledging

- **Promoter Stake**: High stakes (e.g., $> 50\%$) show confidence and control; low stakes (e.g., $< 30\%$) may signal reduced commitment.
- **Pledging Risks**: Promoters pledge shares for loans. High pledging ($> 20\%$ of their holding) can indicate financial distress or risky expansion. If loans default, lenders may sell shares, causing price drops and potential loss of control.
- **Interpretation**: Low or no pledging reflects stability. Rising pledging with high debt is a red flag.

FII - DII & Other Share Holdings

- **FIIs**: Foreign investors (e.g., hedge funds). High stakes ($> 15\%$) show global confidence; increasing stakes signal optimism, but FIIs can be volatile due to global events.
- **DIIs**: Domestic investors (e.g., mutual funds). High stakes ($> 10\%$) indicate stability; DIIs often counterbalance FII sell-offs in India.

- **Interpretation**: Rising FII/DII stakes are positive, but sudden FII exits can cause volatility. DIIs provide long-term stability.
- **Public Shareholding**: High public stakes ($> 25\%$) improve liquidity; low stakes may violate SEBI norms in India (minimum 25% public holding).
- **ESOPs**: Dilute shares but align employee interests.

Example: Reliance Industries Ltd. (RIL) as of March 31, 2025

Table 16: Shareholding Pattern of RIL

Category	Shareholding (%)	Change (%)
Promoters	50.39	-0.10
Promoters' Pledged	5.00 (9.92%)	+2.00
FIIs	24.50	+1.20
DIIs	12.30	-0.50
Public	12.81	-0.60

- **Promoters (50.39%)**: Strong control, minor decrease (-0.10%) likely from ESOPs.
- **Pledging (9.92%)**: Low but rising (+2.00%). Check RIL's debt levels for risks.
- **FIIs (24.50%, +1.20%)**: Growing confidence, positive for price appreciation.
- **DIIs (12.30%, -0.50%)**: Slight decline, possibly profit booking, but stable.
- **Public (12.81%)**: Below SEBI's 25%, may need divestment.

Takeaway: RIL is a strong investment with growing FII confidence, but monitor pledging and public shareholding compliance.

Investor Tips:

- Track pledging trends and debt levels.
- Rising FII/DII stakes are bullish; watch FII exits for volatility.
- Ensure SEBI compliance for public shareholding.

(iii) Share Price Analysis

Share price analysis of a company is crucial for evaluating its past performance and estimating potential returns before making investment decisions. Several key factors and financial indicators should be analyzed to get a complete picture:

Share Price History Analyzing historical share prices helps identify trends, cycles, and anomalies.

- **Growth Rate:** Calculate year-over-year growth using:

$$\text{Growth Rate (\%)} = \frac{\text{Price}_{\text{current}} - \text{Price}_{\text{previous}}}{\text{Price}_{\text{previous}}} \times 100$$

  Positive growth indicates investor confidence or business expansion; negative growth may signal poor performance or market sentiment.
- **Average Annual Growth:** Sum all yearly growth rates and divide by the number of years. Helps evaluate long-term trend.

- **Invested Value Simulation:** If a fixed amount (e.g., ₹10,000) is invested each year, track how many shares are bought annually:

$$\text{No. of Shares}_{\text{year}} = \frac{10,000}{\text{Share Price}_{\text{year}}}$$

  Sum the total shares and multiply with current share price to find total value today.
- **Pattern Recognition:** Look for periods of rapid rise or fall. Link it with known business events such as product launches, crises, mergers, or industry-wide trends.

Dividend History and Analysis Dividends are a major component of shareholder returns, especially for long-term investors.

- **Dividend Per Share (DPS):**

$$\text{DPS} = \frac{\text{Total Dividend Paid}}{\text{No. of Outstanding Shares}}$$

- **Dividend Yield:**

$$\text{Dividend Yield (\%)} = \frac{\text{DPS}}{\text{Market Price per Share}} \times 100$$

  This helps assess income-generating potential of the stock.
- **Dividend Growth Rate:** Calculate year-over-year DPS growth to understand how consistently the company rewards shareholders.
- **Cumulative Dividend Earnings:** For annual investments, compute:

$$\text{Dividend Income}_{\text{year}} = \text{Shares Owned}_{\text{year}} \times \text{DPS}_{\text{year}}$$

  Sum across all years for total passive income generated from dividends.

Return on Investment (ROI) Return on Investment gives a complete view of how much wealth the investment has generated, accounting for both price appreciation and dividends.

$$\text{ROI (\%)} = \left(\frac{\text{Final Value} + \text{Total Dividends} - \text{Total Investment}}{\text{Total Investment}} \right) \times 100$$

Where:

- **Final Value** = Total shares accumulated over years × Current Market Price
- **Total Dividends** = Sum of all yearly dividends earned
- **Total Investment** = ₹10,000 × Number of Years

Negative ROI suggests the stock underperformed or the business faced structural decline. Compare ROI across periods or with similar companies to benchmark performance.

Share Capital and Face Value Analysis Changes in share capital and face value affect the number of outstanding shares.

- **Face Value Adjustment:** If face value reduces (e.g., ₹10 to ₹2), total shares increase proportionally.
- **Dilution Impact:** Higher number of shares may dilute earnings per share unless profits grow accordingly.
- **Interpretation:** Changes in face value usually indicate stock split, which doesn't change value directly but may improve liquidity.

Interpreting Investment Trends From the sample data:

- The share price saw a major decline from ₹544 in 2015 to ₹71 in 2020, a drop of nearly 87%.
- Despite a recovery (CMP ₹123), long-term investors are still in loss if they stayed invested without timely exit.
- The return on ₹10,000 annual investment was strongly negative in later years due to sustained business underperformance.
- Dividend payouts were healthy till 2017 but stopped later, reducing overall investor returns.

Key Insights for Investment Decisions

- Prefer companies with consistent share price growth and dividend history.
- Investigate reasons behind sudden declines (e.g., revenue loss, regulatory issues).
- Diversify investments and regularly monitor business performance.
- ROI and dividend yield must be aligned with your investment horizon and risk profile.

This analysis approach ensures informed and data-driven equity investment decisions.

(iv) Corporate Action Analysis

Corporate actions are pivotal events initiated by a publicly traded company that can significantly impact its shareholders, bondholders, and overall financial structure. These actions, typically approved by the company's board of directors and sometimes requiring shareholder consent, can range from routine dividend payments to complex restructuring events like mergers or demergers. For investors, understanding corporate actions is essential, as they can influence share prices, ownership stakes, and investment returns. This section provides a comprehensive guide to corporate actions, detailing their types, mechanisms, reasons, and impacts on investments, along with guidance on how investors should analyze them.

Basic Concepts

Before exploring corporate actions, it is crucial to understand foundational concepts related to shares and capital, which underpin many of these events.

Face Value

The face value, or par value, of a share is its nominal value as specified in the company's articles of association. Typically a small amount, such as $1 or ₹10, it represents the minimum price at which a share can be issued. While face value has limited practical significance in modern markets, it is critical for accounting purposes and plays a role in corporate actions like stock splits and bonus issues. For example, when a company issues shares, the face value contributes to the calculation of share capital, and changes to face value can affect how corporate actions are structured.

Share Capital

Share capital is the total amount of capital raised by a company through the issuance of shares, calculated as the sum of the face value of all issued shares. It is divided into three categories:

- **Authorized Capital**: The maximum amount of share capital a company is legally permitted to issue, as outlined in its articles of association. For instance, if a company has an authorized capital of ₹1,00,00,000, it can issue shares up to this limit without amending its charter.
- **Issued Capital**: The portion of authorized capital that has been issued to shareholders. For

example, if a company with ₹1,00,00,000 authorized capital issues shares worth ₹50,00,000, the issued capital is ₹50,00,000.

- **Paid-up Capital**: The amount of issued capital that shareholders have paid for. If shareholders have paid for only ₹40,00,000 of the ₹50,00,000 issued capital, the paid-up capital is ₹40,00,000.

These concepts are foundational because corporate actions often involve changes to the share capital structure, such as increasing authorized capital to issue new shares or adjusting issued capital through buybacks.

Types of Corporate Actions

Corporate actions can be classified into mandatory (automatic impact on all shareholders) and voluntary (requiring shareholder participation). Below is a detailed exploration of each type, including their mechanisms, purposes, and impacts on shareholders.

Stock Split

A stock split increases the number of outstanding shares by dividing each existing share into multiple shares. For example, in a 2-for-1 split, each shareholder receives two shares for every one owned, and the price per share is halved.

How It Works The total value of an investor's holdings remains unchanged because the number of shares increases proportionally while the price per share decreases. For instance, if a shareholder owns 100 shares at $200 each (total value $20,000), after a 2-for-1 split, they will own 200 shares at $100 each, still totaling $20,000. The face value is also adjusted proportionally (e.g., from $10 to $5).

Why Companies Do It Companies undertake stock splits to make shares more affordable, attracting a broader investor base, and to increase liquidity and trading volume. A lower share price can enhance marketability, especially for retail investors.

Impact on Shareholders Shareholders receive more shares, but the value per share decreases, maintaining the total investment value. Stock splits are generally viewed positively, as they can signal management's confidence in future growth and increase trading activity. However, multiple splits over time may dilute value if not accompanied by fundamental growth.

Example In 2023, Tesla announced a 3-for-1 stock split, reducing the share price from approximately $660 to $220, making the stock more accessible to investors (`https://www.investopedia.com/articles/03/081303.asp`).

Reverse Stock Split

A reverse stock split consolidates the number of outstanding shares by combining them into fewer shares. For example, in a 1-for-10 reverse split, 10 shares are combined into 1 share, and the price per share increases tenfold.

How It Works The total value of an investor's holdings remains unchanged. For example, if a shareholder owns 1,000 shares at $1 each (total value $1,000), after a 1-for-10 reverse split, they will own 100 shares at $10 each, still totaling $1,000. The face value increases proportionally (e.g., from $1 to $10).

Why Companies Do It Reverse splits are used to boost a low share price, often to avoid being classified as a penny stock or to meet stock exchange listing requirements. A higher share price can improve the company's market perception.

Impact on Shareholders Shareholders own fewer shares, but each share is worth more. While the intrinsic value remains unchanged, reverse splits may reduce liquidity and can be perceived as an artificial attempt to boost the share price, potentially signaling financial distress.

Example In 2022, AMC Entertainment Holdings conducted a 1-for-10 reverse split to increase its share price and comply with exchange listing rules (`https://www.investopedia.com/terms/c/corporateaction.asp`).

Bonus Shares

Bonus shares are additional shares issued to existing shareholders without payment, typically in proportion to their current holdings.

How It Works For example, in a 1-for-1 bonus issue, each shareholder receives one additional share per share owned. This increases the total number of shares outstanding and the issued share capital, often requiring an increase in authorized capital if insufficient. The face value remains unchanged unless combined with a stock split.

Why Companies Do It Bonus shares reward shareholders without distributing cash, often using accumulated reserves. They increase liquidity and make shares more affordable, similar to stock splits.

Impact on Shareholders Shareholders receive more shares, but the value per share may decrease due to dilution. Bonus issues are generally positive, signaling confidence in future earnings, as the company retains cash for growth.

Example In 2023, Larsen & Toubro (L&T) issued bonus shares in a 1:1 ratio, doubling shareholders' holdings without additional cost.

Share Buyback

A share buyback occurs when a company repurchases its own shares from the market, reducing the number of outstanding shares.

How It Works The company uses cash reserves or borrowed funds to buy back shares, which are then canceled or held as treasury stock. This reduces the issued share capital.

Why Companies Do It Buybacks increase earnings per share (EPS) by reducing outstanding shares, signal that the stock is undervalued, and distribute excess cash to shareholders. They can also enhance promoter ownership.

Impact on Shareholders Buybacks can increase EPS and share price, as the company's earnings are spread over fewer shares. Shareholders own a larger percentage of the company, which is generally positive if executed at a fair price.

Example In 2023, Apple announced a $90 billion share buyback program, boosting EPS and signaling confidence in its stock's value (`https://www.investopedia.com/terms/c/corporateaction.asp`).

Dividends

Dividends are payments made to shareholders from a company's profits, either as cash or additional shares.

Types of Dividends

- **Cash Dividends**: Paid in cash, providing direct income.
- **Stock Dividends**: Paid in additional shares, similar to bonus shares.

How It Works Dividends are declared based on profits and board decisions, distributed proportionally to shareholdings. For example, a $0.50 per share dividend yields $50 for 100 shares.

Why Companies Do It Dividends reward shareholders and attract income-seeking investors. They signal financial stability and consistent profitability.

Impact on Shareholders Cash dividends provide income but are taxable. Stock dividends increase shareholdings but may dilute value per share. Dividends are generally positive, though high payouts may indicate limited growth opportunities.

Example In 2023, Reliance Industries declared a dividend of ₹9 per share, providing income to shareholders.

Follow-on Public Offer (FPO)

An FPO is when a listed company issues new shares to the public to raise additional capital.

How It Works New shares are offered through book-building or fixed-price methods. Both existing and new investors can participate, increasing the issued share capital.

Why Companies Do It FPOs raise capital for expansion, debt repayment, or other strategic initiatives.

Impact on Shareholders FPOs dilute existing shareholders' ownership unless they participate. They can be positive if the capital drives growth but may depress share prices due to increased supply.

Example In 2023, Yes Bank conducted an FPO to raise ₹15,000 crore for strengthening its capital base.

Delisting

Delisting occurs when a company's shares are removed from a stock exchange.

Reasons for Delisting

- Acquisition or privatization.
- Failure to meet listing requirements.

How It Works Shareholders may receive a cash offer or shares in another company, depending on the reason for delisting.

Impact on Shareholders Delisting can be positive if shareholders receive a favorable offer or negative if due to poor performance, limiting liquidity.

Example In 2022, Future Retail was delisted after being acquired by Reliance Retail.

Mergers and Acquisitions

Mergers combine two companies into one, while acquisitions involve one company purchasing another.

How It Works In mergers, shareholders of both companies receive shares in the new entity. In acquisitions, shareholders of the acquired company may receive cash or shares of the acquiring company.

Why Companies Do It M&A achieves synergies, expands market share, or enters new markets.

Impact on Shareholders The impact depends on the deal's terms. Successful M&A can enhance value, but integration issues may depress share prices.

Example In 2022, HDFC merged with HDFC Bank, creating a major financial institution.

Demerger

A demerger splits a company into two or more independent entities.

How It Works Shareholders receive shares in the new companies proportional to their holdings in the original company.

Why Companies Do It Demergers focus on core businesses and unlock value in different segments.

Impact on Shareholders Positive if the separate entities perform well, allowing shareholders to benefit from focused operations.

Example In 2023, ITC announced plans to demerge its hotels business.

Spinoff

A spinoff creates a new independent company from an existing division, distributing shares to existing shareholders.

How It Works Shareholders receive shares in the new company without additional payment.

Why Companies Do It Spinoffs allow focus on core businesses and independent growth for the new entity.

Impact on Shareholders Provides investment opportunities in a new entity with potential growth.

Example In 2023, Jio Financial Services was spun off from Reliance Industries.

Rights Issue

A rights issue allows existing shareholders to buy additional shares at a discounted price.

How It Works Shareholders receive rights proportional to their holdings, which can be used to buy new shares or sold to others.

Why Companies Do It Rights issues raise capital with less dilution than public offers, prioritizing existing shareholders.

Impact on Shareholders Allows maintaining ownership proportion; non-participation leads to dilution.

Example In 2023, Hindustan Aeronautics Limited conducted a rights issue to raise ₹2,500 crore.

Liquidation

Liquidation involves selling all assets and distributing proceeds to creditors and shareholders before ceasing operations.

How It Works Assets are sold, debts paid, and remaining proceeds distributed to shareholders.

Why Companies Do It Typically due to bankruptcy or insolvency.

Impact on Shareholders Shareholders are last in line and often receive little or nothing.

Example In 2023, Jet Airways was liquidated after financial struggles.

Priority of Payments on Company Sale

When a company is sold or liquidated, payments follow a priority order:

1. Secured creditors
2. Unsecured creditors
3. Preferred shareholders
4. Common shareholders

Impact on Shareholders Common shareholders are last, often receiving nothing if funds are insufficient.

Name or Symbol Change

A name or ticker symbol change reflects a shift in branding or business focus.

How It Works The company updates its name or ticker, with no direct financial impact.

Why Companies Do It To align with new strategies or improve public perception.

Impact on Shareholders Minimal direct impact but may affect investor perception.

Example In 2022, Facebook changed to Meta, reflecting a focus on the metaverse (`https://investor.fb.com/investor-news/press-release-details/2022/Meta-Platforms-Inc.-to-Change-default.aspx`).

Subsidiary, Alliance, Associate, Joint Venture, Affiliate

These are not corporate actions but structures resulting from strategic decisions:

- **Subsidiary**: Controlled by a parent company.
- **Alliance**: Cooperative agreement between companies.
- **Associate**: Significant influence (20-50% ownership).
- **Joint Venture**: Entity formed for a specific project.
- **Affiliate**: Related through ownership or control.

These structures often arise from corporate actions like M&A or partnerships.

Analyzing Corporate Actions for Investment

Investors should evaluate corporate actions using the following framework:

1. **Reason for the Action**: Determine the company's motive (e.g., rewarding shareholders, raising capital).
2. **Impact on Share Price**: Assess short-term and long-term effects on valuation.

3. **Dilution**: Check if ownership will be diluted (e.g., in FPOs or rights issues).
4. **Financial Health**: Review financial statements to ensure the company can afford the action.
5. **Market Reaction**: Monitor market sentiment and analyst opinions.
6. **Tax Implications**: Understand tax consequences (e.g., dividends are taxable).
7. **Long-term Strategy**: Evaluate alignment with the company's growth plans.

Table 17: Impact of Corporate Actions on Shareholders

Corporate Action	**Primary Impact**	**Investor Considerations**
Stock Split	Increases shares, reduces price per share	Signals growth; assess liquidity impact
Reverse Split	Reduces shares, increases price per share	Check for financial distress; liquidity concerns
Bonus Shares	Increases shares, potential dilution	Evaluate company's reserves and growth prospects
Share Buyback	Reduces shares, increases EPS	Assess buyback price and financial health
Dividends	Provides income or shares	Consider tax implications and payout sustainability
FPO	Dilutes ownership	Evaluate use of proceeds and growth potential
Delisting	Limits liquidity	Assess offer price and reason for delisting
M&A	Changes ownership structure	Review deal terms and synergies
Demerger/Spinoff	Creates new entities	Analyze new entities' growth prospects
Rights Issue	Offers discounted shares	Decide on participation to avoid dilution
Liquidation	Distributes proceeds	Understand priority of payments

5 Sentiment Analysis

5.1 Key Factors Affecting Market Sentiment

1. **Change in Open Interest (OI)**
   Open interest refers to the total number of outstanding derivative contracts (like futures and options) that haven't been settled.
   - Rising OI with price increase suggests new long positions (bullish).
   - Rising OI with price decrease suggests new short positions (bearish).
   - Falling OI with price movement suggests position closure (profit booking).
2. **Delivery Percentage**
   Indicates the portion of traded stocks actually delivered to investors (not just intraday

trading).

- High delivery percentage implies strong investor confidence (bullish sentiment).
- Low delivery indicates speculative or intraday activity (less reliable sentiment).

3. **FII/DII Buying and Selling**
Foreign Institutional Investors (FII) and Domestic Institutional Investors (DII) activity significantly influences market trends.
    - Net buying by FIIs generally boosts sentiment.
    - Net selling indicates bearishness or profit booking.
4. **News**
Market sentiment is heavily influenced by financial, political, and global news.
    - Positive news (mergers, results, economic growth) leads to bullish trends.
    - Negative news (scandals, poor earnings, wars) leads to bearish trends.
5. **Moving Average**
A moving average smooths out price data to identify trends over time.
    - Crossing above a key moving average (e.g., 50-day or 200-day) indicates upward momentum.
    - Crossing below it indicates downward momentum.
6. **Volatility Index (VIX)**
The India VIX measures the market's expectation of volatility over the next 30 days for Nifty.
    - Low VIX implies low fear and stable sentiment.
    - High VIX implies high fear and uncertainty.
    - Generally, **VIX** and **Nifty** move inversely:
        - If VIX $\downarrow \rightarrow$ Nifty $\uparrow$ (bullish)
        - If VIX $\uparrow \rightarrow$ Nifty $\downarrow$ (bearish)
    - If VIX > 22, it indicates high volatility best to avoid fresh investments, especially in derivatives (F&O).
    - $$\text{VIX}/2\sqrt{2}\,\% \rightarrow \text{gives the percentage increase or decrease for next 30 days}$$
7. **Institutional Activity**:
$$\text{FII/DII Ratio} = \frac{\text{Foreign Institutional Investment}}{\text{Domestic Institutional Investment}}$$
    - > 1.2 = Bullish foreign outlook
    - < 0.8 = Domestic confidence
8. **Sentiment Analysis Framework**

Tool	**Bullish Signal**	**Bearish Signal**
Put/Call Ratio	< 0.7	> 1.3
Short Interest	$< 5\%$	$> 20\%$
News Sentiment	70%+ Positive	70%+ Negative
Google Trends	Rising "Buy" searches	Rising "Sell" searches

9. **Research Reports from Reputed Brokerages**
Reports and recommendations from reputed financial institutions influence investor confidence.
    - Positive ratings/upgrades can create bullish sentiment.
    - Negative ratings/downgrades can trigger sell-offs.

10. **Government Policies**
    Policy decisions related to taxation, infrastructure, or regulations have a direct impact on sectors and overall sentiment.
    - Investor-friendly policies = bullish trend.
    - Restrictive or uncertain policies = bearish trend.
11. **Ignore Online Forums**
    Online forums often spread unverified information or hype.
    - Avoid basing trades on unsubstantiated social media discussions.
    - Rely on data-driven or institutional analysis instead.
12. **Investor Interest using Trading Volume**
    Trading volume helps assess investor interest and market strength.
    - High volume with price rise = strong bullish sentiment.
    - High volume with price drop = strong bearish pressure.
13. **Stock Price**
    The price action of a stock reflects real-time sentiment.
    - Consistent price gains show confidence.
    - Declines suggest caution or negativity.

5.2 Price Manipulation Techniques

Stock price manipulation involves artificially inflating or deflating the price of a security to deceive or influence investors.

1. **Increasing/Decreasing Demand or Supply**
   Manipulators can create false buying or selling pressure to move the stock price. For instance, placing large buy orders can increase demand and raise prices temporarily.
2. **Pledging of Shares by Promoters**
   A very high or increasing percentage of promoter-pledged shares may be a red flag. It may lead to price volatility if lenders liquidate shares during price drops.
3. **Low Daily Traded Volume and Delivery Percentage**
   Stocks with low liquidity are easier to manipulate. A small trade can lead to significant price movement, making them vulnerable to artificial price setting.
4. **Volume Changes Without Price Changes**
   Anomalies such as a surge in trading volume without a corresponding price move can indicate accumulation or distribution by manipulators.
5. **Manipulation by High Net-Worth or Retail Investors**
   Even retail investors or HNIs, acting in groups or syndicates, can manipulate illiquid stocks through coordinated buying/selling.
6. **Fundamentally Weak Stocks Are Easier to Manipulate**
   Stocks lacking strong financials or public scrutiny are more prone to manipulation as fewer investors track or question unusual activity.

5.3 Operator Activity

Operators are individuals or entities who artificially drive market prices using tactics such as pump and dump schemes.

- They often form syndicates involving brokers, insiders, or speculators.
- The aim is to manipulate prices for personal gain while appearing as normal market activity.

- They create buying pressure without price movement to accumulate stock, causing others to enter when price starts to rise.
- Once retailers enter, they exit by dumping shares, causing the price to fall.

Typical Activity Pattern

- **Entry:** ↑ Volume → No Price Change → Price likely to rise ↑
- **Exit:** ↑↑ Volume → No Price Change → Price likely to fall ↓

Operator Activity Patterns

Phase	**Volume Pattern**	**Price Action**
Accumulation	Low volume	Sideways movement
Markup	Gradual increase	Steady upward trend
Distribution	High volume	Volatile swings
Dump	Massive volume	Sharp decline

6 Investment Psychology and Portfolio Management

Figure 4: The psychology of Market Cycles

6.1 Behavioral Risks

- **Ego:** Almost everyone tends to believe they are better than average in various aspects of life, particularly in interpersonal skills. This is known as the **"better-than-average effect"**, a well-documented cognitive bias in psychology. In the context of investing, this overconfidence can be particularly dangerous. **Reality Check:** The majority of investors cannot consistently outperform the market. According to efficient market hypothesis (EMH), beating the market requires above-average skill, but by definition, not everyone can be above average. There must be a distribution of returns with some investors performing at or below the average.

- **Coservatism:** refers to the tendency of investors to underreact to new information, remaining anchored to prior beliefs even when the evidence suggests otherwise. This results in delayed or suboptimal decision-making, particularly in fast-changing markets. **Confirmation Bias** further amplifies this issue. Investors often search for, interpret, and recall information in ways that confirm their existing beliefs or hypotheses. For instance, they may focus on the positive aspects of a stock they already own (if long) and ignore warning signs, or vice versa if they are short. The psychological root of these biases lies in our discomfort with cognitive dissonance the mental discomfort we experience when presented with contradictory information. The ability to accept new information, self-reflect, and admit being wrong is rare but essential for successful investing.
- **Attention:** Humans are naturally drawn to vivid narratives, especially during uncertain times. **Attention bias** refers to the tendency to give disproportionate weight to emotionally charged or widely publicized events, while ignoring statistical or probabilistic facts. **Example:** During market crashes, financial news media often dramatizes events with headlines like "Markets in Turmoil" or "Economic Apocalypse". In such times, it's difficult to remember that corrections and bear markets are statistically common, and part of a healthy financial ecosystem. **Behavioral Insight:** This bias is related to the concept of **availability heuristic** where people assess the probability of events based on how easily examples come to mind. Stories and visuals dominate rational thinking.
- **Emotions:** Strong emotions particularly fear and greed are among the most disruptive forces in investing. Emotional decision-making often overrides rational judgment and leads investors to break their own rules. **Example:** An investor who normally sticks to a disciplined asset allocation strategy might panic-sell during a market dip, violating their long-term investment plan.

6.2 Behavioral Biases

- **Forecasting Errors** People overweight recent experiences while making forecasts. This leads to overly optimistic or pessimistic valuations. **Insight:** Investors often extrapolate recent trends into the future, ignoring regression to the mean.
- **Mental Accounting** Investors mentally separate their money into different accounts, each with a different investment style. For example, safe investments for a child's education and risky bets elsewhere. **Problem:** This violates the principle of fungibility of money and can lead to suboptimal portfolio management.
- **Regret Avoidance (Realisation Trap)** Investors feel more regret when unconventional decisions go wrong. They may stick to "mainstream" investments to avoid blame. **Example:** Losing money in a penny stock is more regretful than losing in a blue-chip stock, even if the amount is the same.
- **Loss Aversion Bias** Losses hurt more than gains bring joy. Investors hold onto underperforming stocks hoping to break even, while selling winners too early. **Related Concept:** Average Price Theory averaging down to reduce losses, often leading to poor decision-making.
- **Sunk Cost Fallacy** Investors continue to invest in a losing asset to justify past investments. **Example:** Paying a hefty coaching fee and continuing despite poor quality, just because it's already paid.
- **Decision Paralysis** Too many choices lead to indecision. Investors may delay decisions to avoid regret. **Note:** Not making a decision is itself a decision. Paralysis can be more costly

than making a bad choice.
- **Mental Heuristics** The brain takes shortcuts (heuristics) to process information. Investors simplify complex problems. **Example:** Buying a stock because a few factors seem favorable, ignoring deeper analysis (like buying a stock just because it announces Dividents).
- **False Consensus Effect:** People think that most of the people are thinking the same as they do, but that's not the case. There comes a lot of investment opportunities that come in the eyes of few. Also ways of investment, strategies and choices & hobbies are different for most of the people. Similarly, it's not important that what you are investing in, is invested by the most and vice versa. **Ambiguity Effect:** There are times when we don't know exact result of our decisions, just like investing. So you take views from a lot of people. We should remember that investment strategies are personal and majority opinion is not always right.
- **Herd Mentality and Contrarian Investing** Investors follow the crowd, especially under uncertainty. **Contrarian Wisdom:** "Be fearful when others are greedy, and greedy when others are fearful." **Example:** Contrarian investors buy in bear markets when valuations are low and others are panicking.

6.3 Portfolio Management

Portfolio is a collection of financial investments like Real Estate, stocks, bonds, commodities, cash and cash equivalents, ETFs, mutual funds, etc., and foreign investments.

Types of Risks

- Uncontrollable Risks:
  - Market/Systematic Risk: Uptrend / Downtrend, Macroeconomics
  - Regulatory Changes (restrictions and policies)
- Controllable Risks:
  - Business/Industry Specific (Business cycles, innovations, government regulations)
  - Portfolio Concentration and Optimization

(i) Asset Allocation

Asset Allocation means matching your risk-return profile, goals, and time horizon with your portfolio's assets.

Steps:

1. First establish what is your **risk** and what you expect **returns** from investments.
2. Understand the **goals** you want to conquer and the **time horizon** of your investment.
3. Match your risk with your goals.

Note: Always have some (20–25%) cash to average in future or better invest.

(ii) Diversification

Diversification means investing across different asset classes to maximize your returns and minimize your risks.

“Don’t put all your eggs in one basket”

Diversify your risk-assets properly between high and low risk assets. Even if you are someone who can take risks, it is better to spread your risk and invest across assets that are not related to each other.

Optimal / Proper Diversification To own a number of individual investments large enough to nearly eliminate unsystematic risk but small enough to concentrate on the best opportunities because different industries, sectors, and assets don’t move up and down at the same time or at the same rate.

Statistics: The average standard deviation / risk / volatility level of a single stock portfolio is 49.2%, while increasing the number of stocks in the average well-balanced portfolio could reduce it to a max. of 19.2% (at around 1000 stocks).

- A portfolio of **20 stocks** reduces risk to about 20%.
- The other 980 stocks will reduce it by only 0.8%.

Hence you should not over diversify.

Also, don’t invest more than 25% of your portfolio (in stocks) in a sector.

(iii) Risk Management

Risk Management involves the identification of threats to an outlay and analysing them through standard mathematical approaches or other means, then eventually deriving measures to mitigate the same.

Measures:

1. **Beta or Alpha** of your portfolio
2. **Value at Risk (VaR)** tells you ”How much could I lose in the worst case scenario?” There are certain calculators where you can calculate it easily.

(iv) Rebalancing

Rebalancing involves periodically buying/selling assets in a portfolio to maintain an original or desired level of asset allocation to risk (weighted average risk). It also involves **hedging**. Market and Asset values change all the time. They are affected by a lot of known and unknown factor both domestic and international.
When to consider Portfolio Rebalancing:

- Portfolio directing away from investor’s preference
- Change in your circumstances
- Poor performance of investment
- Global event that shook markets
- Better Opportunities

→ When to Sell Stocks

Sell Triggers

1. Mistake in Buying Decision
2. Stock Hits Target Price
3. Portfolio Rebalancing Needed
4. Personal Financial Requirements
5. Business Fundamentals Deteriorate:
   - Falling Sales/Profits
   - Rising Debt
   - Management Issues
6. Better Investment Opportunities
7. Sector Hype Ends
8. Corporate Governance Issues
9. Dividends Reduced/Eliminated
10. Lower Trading Volumes
11. Price Rises Too Fast Without Reason

7 Other Investing and Stock Market Concepts

7.1 Market Participants

- **Promoters:** Individuals or organizations that help raise funds for a business. They may receive shares or a commission for their efforts.
- **Retail Investors:** Also called individual investors. These are non-professionals who invest in stocks, mutual funds, ETFs, etc.
- **Foreign Institutional Investors (FIIs):** Investment entities registered in a different country that invest in financial markets such as equity, debt, and mutual funds.
- **Foreign Direct Investment (FDI):** Investments made by a company in another country (e.g., setting up a plant or acquiring business assets).
- **Domestic Institutional Investors (DIIs):** Local institutions such as banks, insurance companies, and mutual funds that invest in domestic markets.
- *Note:* Generally, FII (investments) $\propto \dfrac{1}{\text{DII (investments)}}$ though both may sometimes invest or divest simultaneously.
- **Stock Exchange:** Platform where companies raise capital by listing shares for public investment.
  Examples: NSE (National Stock Exchange), BSE (Bombay Stock Exchange).
- **Index:** A statistical measure to track the performance of a group of securities.
  Examples: Sensex, Nifty 50, Dow Jones.
- **SEBI (Securities and Exchange Board of India):** Regulates the Indian capital market.
- **Stock Broker:** A licensed intermediary who executes buy/sell orders and earns via brokerage and maintenance fees.
- **Market Capitalization Categories:**
  - Large Cap: 1st to 100th companies.
  - Mid Cap: 101st to 250th companies.
  - Small Cap: 251st onwards.
- **Mutual Funds:** Investment vehicles that pool money from multiple investors to invest in stocks, bonds, and short-term debt.

7.2 Types of Affiliate Companies

- **Subsidiary:** A company controlled (atleast 50% ownership) by a parent company to gain synergies (e.g., tax benefits, asset control, diversified risk).
- **Associate:** A company in which a parent owns 20-50% stake, allowing participation in policy and strategic decisions.
- **Joint Venture (JV):** A contractual business agreement between two or more entities sharing joint control and pooling resources for the purpose of accomplising a specific task, a new project or any other business activity. *Example Ratios:* 50:50, 60:40, 70:30
- **Affiliates:** Companies related through partial ownership (e.g., less than 50%) or third-party holdings.
- **Mergers:** Two separate companies combine to form a new entity.
- **Acquisitions:** One company takes over another and assumes control.

7.3 Corporate Actions

- **Dividend** is a reward that publicly listed companies extend to their shareholders from the company's net profit. Companies may decide to retain their accumulated profits to reinvest in the business or reserve it for future use or distribute it to shareholders as dividends.
  Announcement about dividends makes a significant increase or decrease in the company's stock value. When a company declares a dividend, its share price undergoes a significant increase and starts to decline by a similar proportion once the date of dividend eligibility expires.
  Important dates: Announcement date, Ex-dividend date, Record date, and Payment date.
- **Bonus Shares** are free shares that the stockholders of the company receive against shares held by them. Bonus shares are issued out of the reserves in shareholders' equity funds. If the ratio is 1:1, the stockholder receives 1 share for every 1 share held.
  In bonus issues, the stock price falls in the same proportion as the bonus issue, so we get no real profits. It is issued as an alternative to dividends or to make the shares affordable.
  Important dates are similar to dividend declaration.
- **Stock Split** is when the number of shares gets multiplied. Technically no new share is issued. If a company announces a stock split in the ratio 1:2, it means for every 1 share held, it will be split into 2 shares.
  Face value of the shares also reduces in the same ratio and so the stock price thus overall no profit. It increases the share liquidity and keeps the share affordable to more shareholders.
- **Reverse Split** is just the reverse of stock split.
- **Rights Issue** is when the company offers new shares to all the existing stockholders in the ratio of their holding in the company at a discount rate to encourage them. This is a primary issue in which the money paid accrues to the company.
  A 1:2 rights issue indicates that stockholders can buy 1 share for every 2 shares held. It's a right (not obligation) and can be traded. The share price also gets reduced in the same proportion. The number of issued shares also increases.
- **Share Buyback** is offered by a company to buy back its shares from the existing stockholders either because share price is too low or because it has surplus capital that it cannot put to good use that it plans to return to the shareholders.
  It decreases the number of issued shares and increases EPS. One more reason can be because company is sure of its future growth. Consider buying if fundamentals are strong, revenue has not been decreased and will have enough liquidity thereafter to issue.
- **Delisting** of a company occurs either by a voluntary decision like below-par performance

of the securities or a merger/acquisition of the listed company with another one or by the SEBI for violation of regulations.
It will require shareholder approval and buys them on a price bid by the shareholders, but it must be also acceptable to the company.

- **Priority of Payments Upon Company Sale:** When a company is sold, the proceeds from the sale are distributed in a specific order to fulfill the company's financial obligations. The typical priority of payments is as follows:
  1. **Interest to Lenders:** Payments are first made to lenders, including those holding debt, bonds, and debentures.
  2. **Taxes to Government:** Next, tax liabilities owed to the government are settled.
  3. **Dividends to Preferred Shareholders:** If there are any preferred shareholders, they are paid their fixed dividends after taxes are cleared.
  4. **Reserves and Surplus to Common Equity Shareholders:** Finally, any remaining surplus is distributed to common equity shareholders, often referred to as retained earnings or reserves and surplus.